Scattering Characteristic-Aware Fully Polarized SAR Ship Detection Network Based on a Four-Component Decomposition Model

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***Abstract*— Model-based decomposition methods are widely used in full-polarization synthetic aperture radar (SAR), for the inversion and interpretation of ground features and constitute an important approach for understanding the behavior of backscattering. However, due to the substantial differences between land and marine environments, different man-made and natural vegetation scattering structures render existing decomposition models unable to reasonably characterize scatterers on ships. Moreover, the combination of polarization decomposition models and neural networks for ship detection has rarely been investigated. Therefore, this study proposes a four-component decomposition model (Ship-4SD) suitable for describing the scattering characteristics of ships based on the surface scattering, double-bounce scattering, ±45° oriented dipole, and asymmetric scattering components. Furthermore, based on the differences in the scattering properties exhibited by different scattering components in ships and the powerful feature extraction capability of convolutional neural networks (CNNs), a scattering characteristic-aware fully polarized SAR ship detection network (SCANet) was designed to make full use of the scattering components in the decomposition model. Finally, the experimental results on a large amount of GF-3 fully polarized SAR (PolSAR) data validated the reasonability and superiority of Ship-4SD and SCANet. The Ship-4SD can better distinguish ship and clutter pixels compared to other four-component models and has a higher target-clutter ratio with respect to the multicomponent models. SCANet proposed in this article achieved an average precision (AP) of 94.43% and 96.56% on the GF-3 and SSDD datasets, respectively, which is better than that of other competitive CNN algorithms.**

***Index Terms*— Physical scattering model, scattering characteristic awareness, ship detection, ship scattering power decomposition.**

I. Introduction

SYNTHETIC aperture radar (SAR) is extensively used in military and civil applications due to its unique advantages of all-day and all-weather continuous observation [1], [2]. Maritime target detection is an essential component of Earth observation research as accurate ship positioning information can help governments manage maritime emergencies [3]. The backscattering information of polarized SAR (PolSAR) contains rich target polarization characteristics that can reflect the geometric structure and physical properties of the target, helping completely characterize its scattering properties [4]. Thus, ship detection using PolSAR data is currently a widely used method. Among the available ship detection algorithms, the model-based decomposition of polarization scattering characteristics has good prospects for application because it has strong physical interpretability and can represent the average mechanism of the target scattering for each resolution unit. This separates the scattering mechanisms presented by different components in the target and reveals the essential differences between the target and sea clutter scattering [5].

The decomposition objects of the model-based decomposition algorithm for the polarized scattering properties include the scattering [6], Kennaugh [7], and covariance or coherence matrices [8]. Among these, covariance or coherence matrix-based decomposition algorithms are the most widely used. They typically assume that scatterers exist in a scattering target. For example, Freeman and Durden [9] assumed the existence of reflection symmetry in the scattering mechanism of the natural distribution region and proposed a three-component model (FDD). However, terrestrial areas (such as urban areas) do not always satisfy reflection symmetry; therefore, it is necessary to consider nonreflection symmetry. Based on these results, Yamaguchi et al. [10] extended the FDD to a four-component model (Y4O) by adding a helix scattering (Hlx) component to the FDD. This four-component decomposition model is effective for both urban and vegetated areas. However, Y4O contains some inherent defects, such as orientation dependence [11], indistinguishable urban and vegetation [12], overestimation of volume scattering power [13], and underutilization of covariance matrix components [14]. Therefore, subsequent research was conducted to address these limitations. Yamaguchi et al. [11] further proposed the orientation-independent four-component decomposition model (Y4R). Sato et al. [12] proposed a four-component decomposition algorithm (S4R) with an extended volume-scattering model which led to improved identification in vegetation and urban areas. Bhattacharya et al. [13] proposed a method (BF4) to estimate the orientation angle (OA) [15] of fully PolSAR images using the Hellinger distance and power correction of the Y4O decomposition model. Singh et al. [14] proposed a new generalized four-component decomposition

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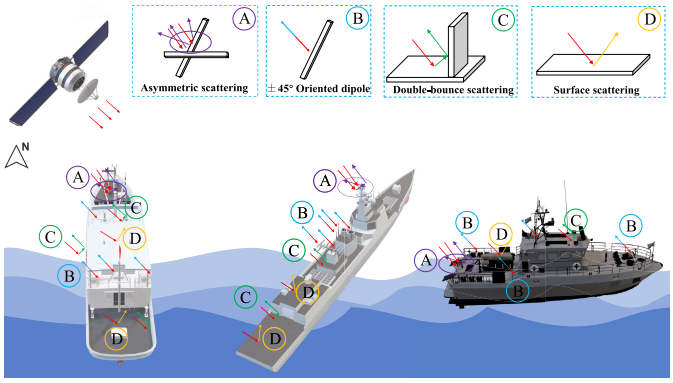


Fig. 1. Ship backscattering characteristics analysis.

model (G4U) by performing a set of unitary transformations on the polarization coherence matrix. The G4U reduces the number of independent parameters from eight to seven and makes full use of the polarization information in the coherence matrix to improve the double-bounce reflected power in urban areas. In addition to the four-component decomposition model, other multicomponent models have been proposed, such as the multiple-component scattering model (MCSM) [5], six-component scattering matrix power decomposition (6SD) [16], and seven-component scattering matrix power decomposition (7SD) [17]. However, the complexity of the model increases with the number of components. Therefore, this study compromised on the complexity and efficiency of the decomposition model to focus on the four-component decomposition model.

The applications of existing model-based decomposition algorithms are currently predominantly to terrestrial vegetation and urban areas while those to the ships are less studied. Existing models typically use surface scattering (Odd), double-bounce scattering (Dbl), volume scattering (Vol), and Hlx components to describe flat plates, dihedral structures, dense canopy regions, and the nonsymmetric scattering properties of features, respectively. In fact, the flat and dihedral structures described by the Odd and Dbl components are widely present in natural vegetation and man-made features and thus can be applied to a variety of scenarios, including ships at sea. However, the Vol component describes a dipole cloud, similar to a tree canopy. The corresponding scattering structures that rarely exist on ships composed of metallic materials, such as steel and metal parts (e.g., masts and railings on the ship's deck), are linear steel frame structures. Moreover, the Hlx component cannot completely describe the asymmetric scattering characteristics of ships, due to the complexity of ship construction. Overall, existing scattering models are unable to reasonably characterize ships.

This implies the urgent need to develop a new decomposition model to reasonably describe the scattering structure of a ship. Fig. 1 shows three representative ships with a complex scattering structure from different viewpoints (we give several representative angles including 0°, 45°, and 90° between the direction of motion and the direction north as examples). When the image resolution is sufficient for ship detection, the scattering structures on a ship can be identified from different prospectives corresponding to the four scattering components: Odd, Dbl, ±45° oriented dipole (Od), and asymmetric scattering (Asym). Thus, in addition to Odd and Dbl, Od and Asym should be introduced to describe the scattering characteristics of ships. Additional reasonings for the above propositions are as follows.

1) The Od component can describe the composite scattering structure consisting of even-bounce and odd-bounce reflectors. The upright metal parts (even-bounce reflector) on the ship deck can form a composite scattering structure with the deck (odd-bounce reflector). As analyzed in [16], these composite scatterring structures can be represented by a scattering matrix of a ±45° oriented dipole component.
2) The Asym component is used to describe the asymmetric scattering characteristics of a ship. So far, an attempt has been made to characterize the asymmetric scattering mechanism using helix scattering. However, the helix scattering fails to utilize the three quantities related to asymmetric scattering, thus failing to provide a complete description of the asymmetric scattering properties of an object [18]. Therefore, an asymmetric scattering component needs to be introduced to overcome the limitations of the helix scattering mechanism and characterize the complete asymmetric scattering information of the ship.

In summary, to make the model-based decomposition algorithm more suitable for ship detection, we propose a model-based four-component decomposition algorithm for maritime ships using the Odd, Dbl, Od, and Asym components to describe the complete scattering characteristics of a ship.

In addition to the decomposition model, the use of these scattering components for ship detection remains an issue that requires further investigation. In recent years, convolutional neural networks (CNNs) have been widely applied to SAR image ship detections due to their powerful feature extraction capabilities. For example, Li et al. [19] employed an enhanced and faster R-CNN network for ship detection in SAR images and released an SSDD dataset. Lin et al. [20] modified the faster R-CNN network structure for SAR image ship detection using a squeeze excitation mechanism and verified the superiority of the network using Sentinel-1 images. Cui et al. [21] constructed a large-scale SAR image target detection network by introducing a spatial shuffle-group enhancement module into the CenterNet. To solve the problem of insufficient ship detection performance in nearshore scenarios, Wang et al. [22] proposed a semisoft label-guided SAR ship detection network based on self-distillation.

Notably, the above CNN detection methods have achieved good detection performance on publicly available SAR intensity datasets such as the SSDD [23], LS_SSDD [24], and OpenSARShip [25]. However, it should be noted that research using CNN methods focuses more on the model structure using intensity values alone. Studies that use CNN to detect PolSAR ships are rare [26]. Because full PolSAR data contain rich backscattering information, conducting research on CNN methods for PolSAR data is an urgent future task.

Based on previous considerations, this study proposes a scattering characteristic-aware fully PolSAR ship detection method. The contributions of this study can be summarized as follows.

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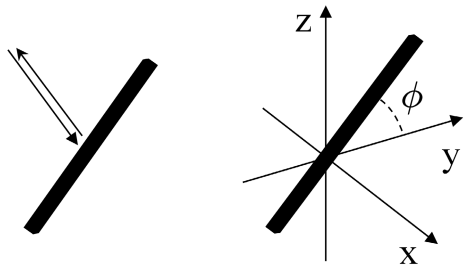


Fig. 2. $\pm 45°$ oriented dipole model, $\phi = 45°$.

1) A ship-scattering four-component decomposition (Ship-4SD) model applicable for characterizing ship scattering has been proposed. By analyzing the scattering characteristics of ship structures and profiling the shortcomings of the Vol and Hlx components applied to ships, we employed the Od component to describe composite structures consisting of flat plates (odd-bounce reflectors) and $\pm 45°$ oriented dihedral (even-bounce reflectors) on ships. All asymmetric scattering structures in the ship were characterized using Asym. Consequently, Ship-4SD is derived from both mathematical and theoretical perspectives.

2) A scattering characteristic-aware fully PolSAR ship detection network (SCANet) applicable to Ship-4SD is proposed. SCANet can automatically perceive the contribution weights of different scattering mechanisms represented by Ship-4SD to serve PolSAR ship detection, which provides a way of applying scattering components based on model decomposition to target detection.

3) Experiments on a large amount of measured fully PolSAR data to validate the reasonability and effectiveness of Ship-4SD and SCANet.

The remainder of this article is organized as follows. Section II presents a derivation of the proposed Ship-4SD. The structure of SCANet is described in detail in Section III. In Section IV, the four-component model is validated using GF-3 fully PolSAR data, and SCANet is tested on the GF-3 and SSDD datasets. Finally, Section V summarizes the study and its major results.

II. Four-Component Decomposition Model for Ships

The Ship-4SD model proposed in this study comprises the Odd, Dbl, Od, and Asym components. In this section, we analyze four basic types of ship scatterers, as shown in Fig. 1.

A. Basic Scattering Components

According to the basic theory of target decomposition, all the scattering characteristics of an object can be represented by a scattering $[S]$ [27] or coherence matrix $[T]$ [16]. $[S]$ and $[T]$ are expressed as follows:

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}, \quad [T] = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \tag{1}$$

where S_{pq} denotes the electromagnetic waves sent to p and received at q. The horizontally and vertically polarized waves are, respectively, by H and V.

For a ship, the different scattering components can be expressed by

1) Odd:

$$[S]_s^{\text{total}} = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$[T]_s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{modified}} [T]_s = \begin{bmatrix} 1 & \beta^* & 0 \\ \beta & |\beta|^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{2}$$

where $[S]_s^{\text{total}}$ and $[T]_s$ are surface scattering submatrices. The derivation of $[T]_s$ is explained in [28]. β is the parameter to be solved. $*$ denotes the conjugate transpose. $|\cdot|$ indicates the modulus of the complex signal.

2) Dbl:

$$[S]_d^{\text{total}} = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$
$$[T]_d = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{modified}} [T]_d = \begin{bmatrix} |\alpha|^2 & \alpha & 0 \\ \alpha^* & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{3}$$

where $[S]_d^{\text{total}}$ and $[T]_d$ are double-bounce scattering submatrices. The derivation of $[T]_d$ has been described in [28]. α is the coefficient to be solved.

3) Od: The scattering characteristics of an oriented dipole significantly depend on the target orientation with respect to the polarization coordinate system. Fig. 2 shows the observed behavior when there is a dipole in the target pointing at $\pm 45°$. By combining the scattering matrix of an odd-bounce reflector (flat plate) with that of a $\pm 45°$ oriented dihedral, a composite-oriented dipole scattering matrix was formed [16]

if $\Re(T_{13}) >= 0$

$$[S]_{\text{od}}^{45°} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad [T]_{\text{od}}^{45°} = \frac{1}{2}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \tag{4}$$

if $\Re(T_{13}) < 0$

$$[S]_{\text{od}}^{-45°} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad [T]_{\text{od}}^{-45°} = \frac{1}{2}\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} \tag{5}$$

where $\Re$ is the real part. $[S]_{\text{od}}^{45°}$, $[T]_{\text{od}}^{45°}$, $[S]_{\text{od}}^{-45°}$, and $[T]_{\text{od}}^{-45°}$ are $\pm 45°$ oriented dipole submatrices. The derivations of $[T]_{\text{od}}^{45°}$ and $[T]_{\text{od}}^{-45°}$ are presented in [16].

In practice, ships at sea and buildings on land are similar. For example, both building structures steel (bridges and streets) and ship ones (decks, containers, and masts) can form composite scattering structures based on flat-plate scattering and directional dihedral structures. As indicated in [16], the Od component can adequately describe such scattering structures. Therefore, it is reasonable to use the Od component to characterize the composite scattering structure of a ship.

4) Asym: Existing four-component decomposition models assume that the probability density function of the rotation angle around the radar line of sight is uniformly distributed, from which it can be deduced that $T_{13} = T_{23}$ [10]. The imaginary part of T_{23} is then modeled using the Hlx component to describe the nonreflective symmetry of many urban

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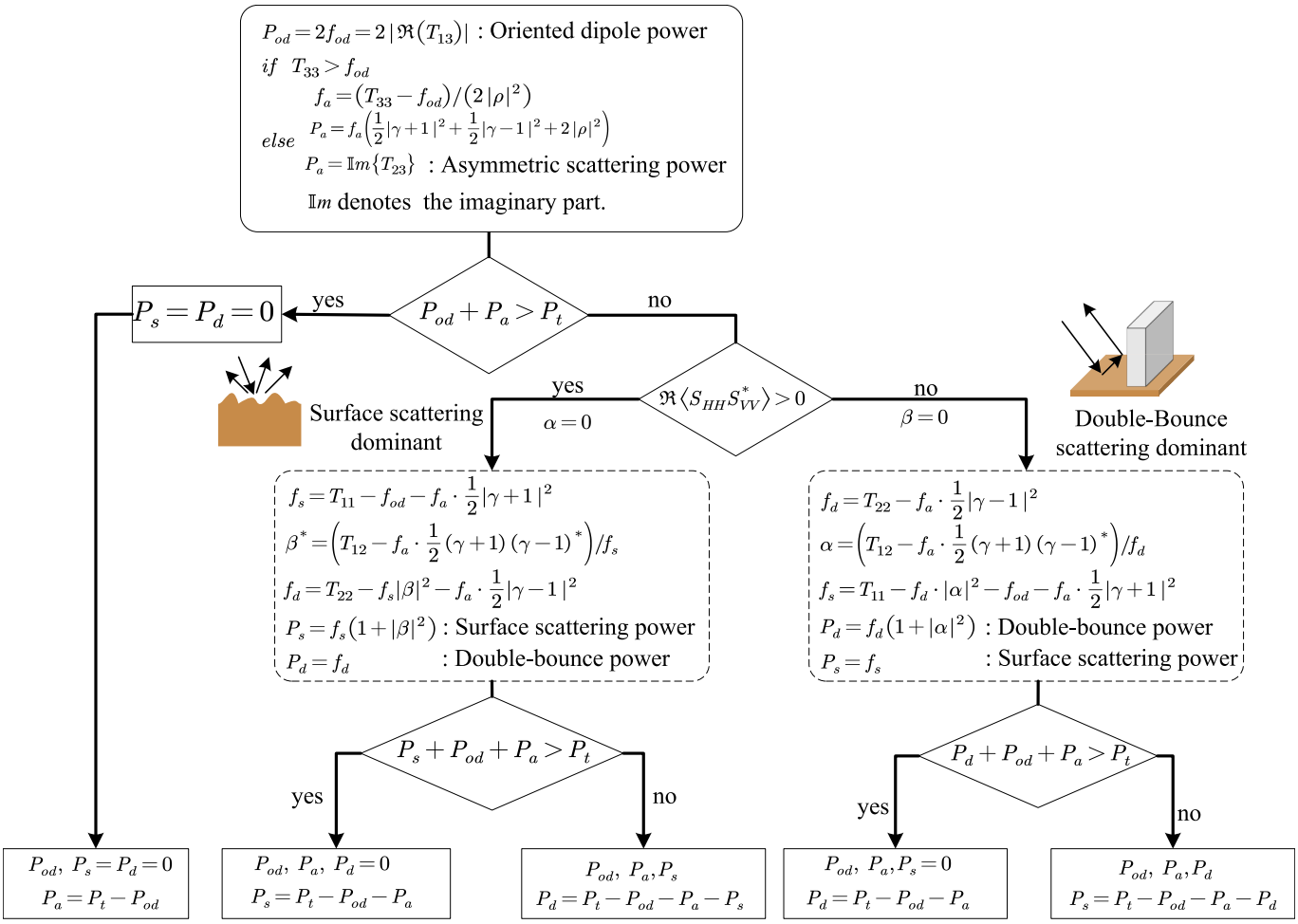


Fig. 3. Ship-4SD decomposition flowchart.

areas. However, the actual measured data suggests that this assumption does not always hold true; that is, T_{13} and T_{23} are not always equal to purely imaginary numbers. This implies that the existing scattering models for these areas do not agree with the real data. As pointed out by [18], it is not possible to describe all asymmetric scattering in urban buildings using only helix scattering (indicated by the Hlx component). In addition, some studies have shown that other fundamental scattering mechanisms, apart from helix scattering, have nonreflective symmetric scattering properties [5], [15], [29].

Therefore, based on the above analysis, Asym components are introduced to better describe the asymmetric scattering structures contained in ships. It is known that T_{13} and T_{23} are parameters related to the nonreflective symmetric scattering information [18]. In previous four-component decomposition models, only the imaginary part of T_{23} was used to represent the nonreflective symmetry of the features; the remaining elements were not used. We expect the Asym component to consider all parameters related to asymmetric scattering information on a ship; that is, the Asym component should have all T_{13} and T_{23} terms. However, due to the complexity of the asymmetry, there is no exact mathematical model for describing the Asym component containing both T_{13} and T_{23}. The general form of the Asym component is

$$[S]_a=\begin{bmatrix}\gamma & \rho\\ \rho & 1\end{bmatrix} \tag{6}$$

where γ and ρ denote the ratios of the backscattering coefficients of HH and HV to those of the VV polarization channel [18]. Based on the $[S]_a$, the Pauli scattering vector k_a can be writed by $k_a=(1/(2)^{1/2})[\gamma+1,\gamma-1,2\rho]$.

The coherence matrix of the Asym component is expressed as follows:

$$[T]_a=\langle k_a k_a^*\rangle. \tag{7}$$

From (7), it can be seen that $\langle(\gamma+1)\rho^*\rangle$ and $\langle(\gamma-1)\rho^*\rangle$ are complex terms corresponding to T_{13} and T_{23}, respectively. Therefore, the Asym component can describe the nonreflective symmetric properties presented by asymmetric scatterers and overcome the limitations of Hlx. $\langle\cdot\rangle$ represents ensemble averaging.

B. Decomposition

A four-component decomposition model based on the coherence matrix can be constructed using the Odd, Dbl, Od, and

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Asym components to describe the scattering structures on the ship. The model can be expressed as

$$[T] = f_s[T]_s + f_d[T]_d + f_{\text{od}}[T]_{\text{od}} + f_a[T]_a \tag{8}$$

where f_s, f_d, f_{od}, and f_a denote the four unknown coefficients to be determined, based on the following equation:

$$\begin{cases} T_{11} = f_s + f_d \cdot |\alpha|^2 + f_{\text{od}} + f_a \cdot \dfrac{1}{2}|\gamma + 1|^2 \\ T_{12} = f_s \cdot \beta^* + f_d \cdot \alpha + f_a \cdot \dfrac{1}{2}(\gamma + 1)(\gamma - 1)^* \\ T_{13} = f_{\text{od}} + f_a \cdot (\gamma + 1)\rho^* \\ T_{22} = f_s \cdot |\beta|^2 + f_d + f_a \cdot \dfrac{1}{2}|\gamma - 1|^2 \\ T_{23} = f_a \cdot (\gamma - 1)\rho^* \\ T_{33} = f_{\text{od}} + f_a \cdot 2|\rho|^2. \end{cases} \tag{9}$$

In the above model, γ and ρ are known. Consequently, there are six parameters to be solved: f_s, f_d, f_{od}, f_a, α, and β. According to [16], we can directly obtain the power of Od

$$P_{\text{od}} = 2|\Re(T_{13})|, \quad P_{\text{od}} = 2\, f_{\text{od}}. \tag{10}$$

Therefore, f_{od} is solved first.

Then, using the last equation in 9 and based on f_{od}, f_a can be solved for. Subsequently, the remaining four unknown parameters are referred to in [11] and [16].

Finally, the power values P_s, P_d, and P_a for the Odd, Dbl, and Asym components are expressed as follows:

$$\begin{aligned} P_s &= f_s\left(1 + |\beta|^2\right) \\ P_d &= f_d\left(1 + |\alpha|^2\right) \\ P_a &= f_a\left(\frac{1}{2}|\gamma + 1|^2 + \frac{1}{2}|\gamma - 1|^2 + 2|\rho|^2\right). \end{aligned} \tag{11}$$

Therefore, the total power $P_t = P_s + P_d + P_{\text{od}} + P_a$. Fig. 3 shows the Ship-4SD decomposition flowchart. It should be noted that when solving for P_s and P_d, four unknown parameters were included in the three equations, making the equations unsolvable. Thus, we used the positive and negative relationship of $\Re\langle S_{HH} S_{VV}^* \rangle$ according to the solution in [11], [16] to determine whether the Od or Dbl components prevail, and then set $\alpha = 0$ or $\beta = 0$ such that the four unknown quantities become three; hence, P_s and P_d can be solved. Unfortunately, this operation may result in a negative-power problem for P_s or P_d. Similarly, we adopted the treatment in [11] and [16], where $P_s = 0$ or $P_d = 0$ was set to maintain a constant total power when P_s or P_d was negative, subject to the constraint of total power P_t. The rationality of these settings is confirmed in Section IV.

III. Scattering Characteristic Awareness Network for Fully PolSAR Ship Detection

The structure of SCANet is shown in Fig. 4. SCANet includes the backbone network ResNet50 [30], scattering characteristic awareness module (SCAM), and feature pyramid network (FPN). The input-polarized scattering features were first mined using ResNet50, which outputs three feature maps at different scales, S3, S4, and S5, as shown in Fig. 4. Then, S3, S4, and S5 were input into the SCAM. The SCAM contains an efficient pyramid split attention block (EPSAB) and a receptive field block (RFB), which were used to mine the scattering features of a ship under different scattering mechanisms and the local and global features of the ship, respectively. Finally, the feature outputs from the SCAM were fed into the FPN for feature fusion. It is worth noting that selecting ResNet50 is because it is a widely used backbone network with residual learning block. This block can solve the problem of vanishing/exploding gradients due to deep network depth, thus deepening the network depth and improving the network performance [31]. Moreover, an FPN is commonly used for multiscale target detection, which matches multiscale ships in SAR images [32]. This section describes the proposed SCAM.

A. Details in SCAM

In Section II, a four-component decomposition model suitable for characterizing ship scattering was derived. As we know, different scattering mechanisms have different abilities to characterize ships and backgrounds [33]. For the components in Ship-4SD, Odd is not the main scattering mechanism on ships and is more suitable for describing the sea surface. Instead, the main scattering mechanisms in ships are the Dbl, Od, and Asym components. Unfortunately, it is challenging to understand the contributions of the different scattering mechanisms.

1) EPSAB: The visual attention mechanism is a special brain signal-processing system for processing visual information [34]. Human vision swiftly scans an entire image to identify the area that needs to be concentrated, that is, the focus of attention. It then applies more attentional resources to the focused target area to gather more comprehensive information about the target and suppress irrelevant information. Deep learning has borrowed this human information-processing mechanism and applied it to various applications in image and natural language processing, achieving breakthrough results. Thus, using the visual attention mechanism to mine the scattering characteristics of targets under different scattering mechanisms is a good way to improve the target detection accuracy and reduce false alarms.

Therefore, inspired by the attention mechanism and combined with the characteristics of the input data in this study, we introduced EPSAB in the SCAM, whose structure is shown in Fig. 5. The four main steps of EPSAB are as follows.

Step 1: Split the feature map, extract multiscale features of the target on each channel feature map, and concatenate all multiscale feature maps. This step has the advantage of effectively extracting the spatial information exhibited by ship targets with different scattering characteristics using the following expressions:

$$\begin{aligned} G &= 2^{\frac{k-1}{2}} \\ F_i &= \text{Conv}(k_i \times k_i, G_i)(X_i) \\ X &= \text{Concat}\left[X'_1, X'_2, \ldots, X'_{S-1}\right] \end{aligned} \tag{12}$$

where multiscale feature extraction is accomplished by the group convolution of multiscale convolution kernels, $i = 0, 1, 2, \ldots, S-1$, S denotes the number of groups to be grouped,

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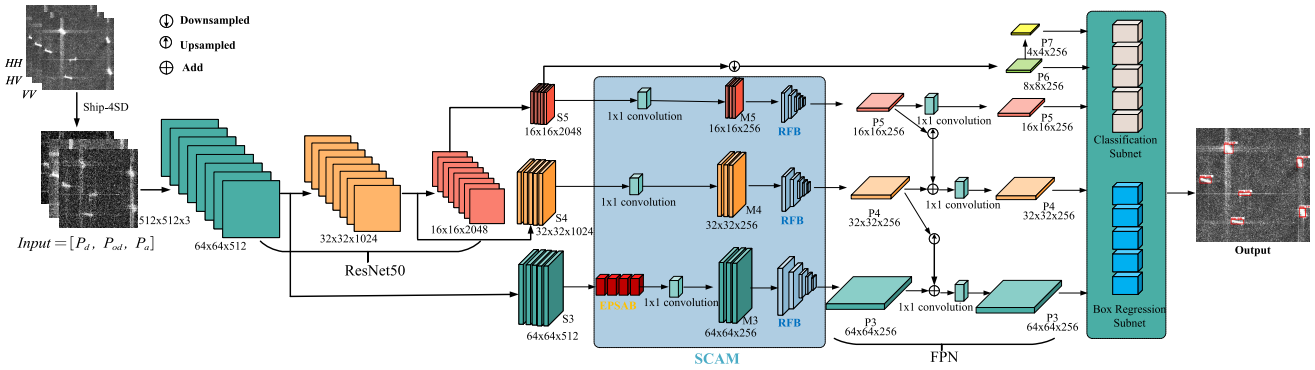


Fig. 4. SCANet structure.

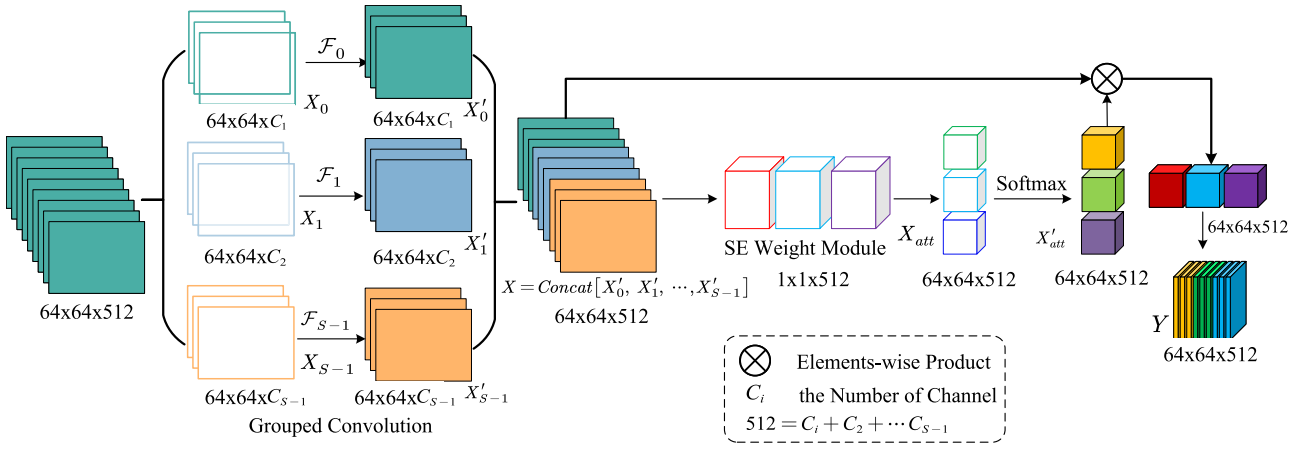


Fig. 5. EPSAB structure.

and G is the number of groups. k is the kernel size and X_i denotes the grouped feature map. X_i' indicates the feature maps at different scales, and X is the concatenated result of the feature maps at different scales. Conv denotes convolution. Concat indicates that feature maps are connected through channels.

Step 2: Use the SE weight module [35] to extract the channel attention weights of the feature maps at different scales. The SE Weight module can be expressed as

$$X_{\text{att}} = \text{SEWeight}(X). \tag{13}$$

The attention weights for the cth channel in the SE module can be written as

$$w_c = \sigma(W_1\delta(W_0(g_c))) \tag{14}$$

where σ denotes the excitation function; W_0 and W_1 indicate the fully connected layers; δ indicates a rectified linear unit (ReLU) operation; and g_c denotes the global average pooling operator. The weight can recognize the contribution of different channels to the ship's target.

Step 3: Use the Softmax function to extract features and rescale the multiscale channel attention vector to obtain the attention weights of all channels of the new multiscale feature map

$$X'_{\text{att}} = \text{Soft}\max(X_{\text{att}}). \tag{15}$$

Step 4: Channel-wise multiplication of new weights and feature maps (which are not processed by the SE weight module). Consequently, we can obtain a multiscale feature map after attention weighting of the feature information

$$Y = X \otimes X'_{\text{att}} \tag{16}$$

where the $\otimes$ indicates channel-wise multiplication.

In the SCAM, we input the $64 \times 64 \times 512$ shallow feature map S3 into EPSAB. The advantage of this strategy is that the shallow feature map contains rich and detailed information regarding the ship's target, such as its outline and shape. Inputting S3 into the EPSAB makes maximum use of the detailed information of the shallow feature map, which is beneficial for the detection of small targets. In addition, after EPSAB, we reduced the dimensionality of the feature maps using a 1×1 convolution on S3, S4, and S5, which helped reduce the amount of data and enabled information interaction between different channels.

2) Receptive Field Block: To fuse the ship features contained in different layers, the RFB was added to the SCAM with four different dilation rates in the four branches for feature extraction of the input feature maps. Fig. 6 shows the dilation rates used in this article and the code flow. The feature maps S3, S4, and S5 were then input into the RFB to improve the ability of the network to retrieve features by replicating the human visual perceptual field. Fig. 7 shows the receptive fields of the convolution kernel at different expansion rates.

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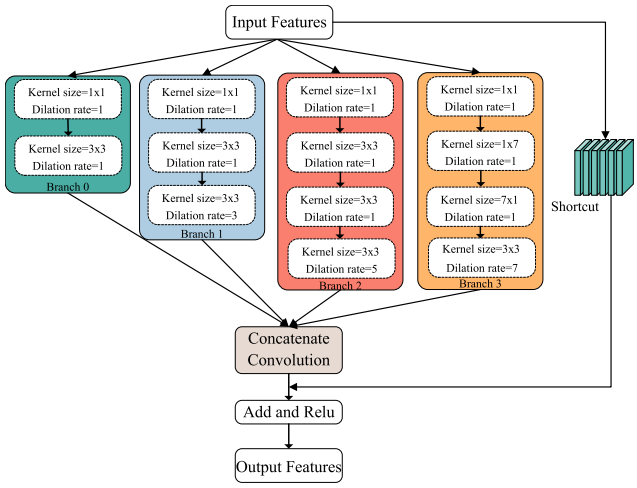


Fig. 6. RFB structure.

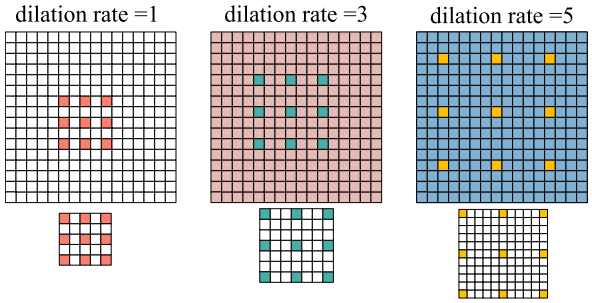


Fig. 7. Dilation convolution.

From Fig. 7, it can be seen that the receptive field is enlarged when the dilation rates are set to 1, 3, 5 with the same size of the convolution kernel. Therefore, a convolutional kernel with a large expansion can capture a larger range of features without increasing the size of the convolutional kernel, facilitating the network's judgment of targets in the scene.

The relationship between the convolution kernel and the receptive field can be expressed as

$$r = k + (k-1)(\epsilon - 1) \tag{17}$$

where r is the size of the receptive field, k is the convolution kernel, and ϵ is the dilation rate. The relationship between the input and output feature maps for different receptive field sizes can be expressed as

$$O = (I - r + 2p)/s + 1 \tag{18}$$

where O, I, and p denote the output feature map, input feature map, and padding size, respectively, and s denotes the step size.

B. Loss Function

For target detection, the loss function consists of two main aspects: classification and regression losses. In single-stage target detection networks, detection accuracy is often low. The reason is that on the one hand, the number of negative samples is too large, resulting in the loss of positive samples being covered; on the other hand, hard samples are often the transition between the foreground and background areas, and these samples are few and difficult to distinguish. These two factors were reflected in the model, resulting in poor performance and detection accuracy. To solve this problem, Lin et al. [36] proposed a focal loss method to quantify classification loss

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\eta} \log(p_t) \tag{19}$$

where $\eta \in [0, 5]$ denotes the frequency modulation factor and $\alpha_t \in [0, 1]$ is the weighting factor. p_t denotes sample confidence.

In this study, we used the smooth_{L1} function to calculate the regression loss [37]. This loss function is typically used for object detection [38]

$$\mathrm{smooth}_{L1} = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{20}$$

where x denotes the localization offset between the predicted and ground truth boxes. In summary, the total loss function is

$$\mathrm{Loss}(p_i, t_i) = \frac{1}{N_{\mathrm{cls}}} \sum_i \mathrm{FL}\left(p_i, p_i'\right) + \lambda \frac{1}{N_{\mathrm{reg}}} \sum_i \mathrm{smooth}_{L1}\left(t_i, t_i'\right) \tag{21}$$

where p_i and p_i' are the true class label and prediction score, respectively. t_i and t_i' denote the ground-truth and predicted box coordinates, respectively. λ denotes the weight coefficient of regression loss. N_{cls} and N_{reg} denote the batch sizes during training.

C. Data Input

As analyzed in Section III-A, there are large differences in the texture characteristics, and geometry, among others, presented by the ship under different scattering mechanisms, because the sea surface mainly presents a surface scattering mechanism, and the structure of the ship cannot be described under the same mechanism. Therefore, we discarded the Odd scattering characteristic data and used the Dbl, Od, and Asym components as inputs to the network to drive the SCANet with the following expression:

$$\mathrm{Input} = [P_d, P_{\mathrm{od}}, P_a]_{512\times512}. \tag{22}$$

IV. Experimental Results and Discussions

A. Experiment Design and Dataset

The experiments were divided into two parts. The first part verified the rationality and effectiveness of Ship-4SD. Five model-based decomposition algorithms for Y4O, Y4R, S4R, G4U, and BF4 and two multicomponent decomposition algorithms (6SD and 7SD) were compared with Ship-4SD using large amounts of GF-3 SAR data. Because of the large amount of data in GF-3, we list only the information of the data used for visualization in this study in Table I. The AIS information corresponding to all the SAR data was made available by the National Satellite Ocean Application Service of China. The details of the data are presented in Table I.

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TABLE I
GF-3 DATA INFORMATION

No.	Product ID	Imaging Area	Image Size	Date	Mode	Incidence Angle	Resolution (Azimuth x Range)
1	5663989	Yangtze Estuary	5853*6794	2021/05/24	QPSI	44.90∼45.94	8m*8m
2	4128590	Qinhuangdao	8752*7038	2021/07/19	QPSI	23.66∼26.34	8m*8m
3	4056567	Gulf of Oman	6484*8062	2019/06/13	QPSI	35.61∼37.42	8m*8m
4	3564287	South Bay	8082*7882	2018/10/31	QPSI	31.08∼33.23	8m*8m
5	5691180	Huangqi Gulf	6356*3998	2021/06/09	QPSI	46.16∼47.31	8m*8m
6	5691179	Minjiang Estuary	6356*3977	2021/06/09	QPSI	46.16∼47.31	8m*8m
7	4008613	Osaka Port	6486*8062	2019/05/19	QPSI	35.49∼37.30	8m*8m
8	6154474	Bohai Gulf	7675*6014	2021/12/20	QPSI	19.87∼22.67	8m*8m
9	6154462	the Yellow Sea	7675*6014	2021/12/20	QPSI	20.04∼22.82	8m*8m
10	2599253	Half Moon Bay	7173*5829	2021/09/15	QPSI	19.86∼22.58	8m*8m
11	6154442	Port of Refuse	7675*6014	2021/12/20	QPSI	20.27∼23.02	8m*8m
12	6154473	Bohai Sea	7675*6014	2021/12/20	QPSI	19.89∼22.68	8m*8m
13	4128588	Bohai Gulf	8752*7038	2019/07/19	QPSI	23.64∼26.33	8m*8m
14	4128582	Yellow River Port	8752*7038	2019/07/19	QPSI	23.69∼26.37	8m*8m
15	6191496	the Yellow Sea	5972*4288	2022/01/03	QPSI	46.92∼48.11	8m*8m
16	4136813	Ikata town	6485*8062	2019/07/23	QPSI	35.51∼37.33	8m*8m
17	4138161	Huangjiatang Bay	4707*6014	2019/07/24	QPSI	38.55∼39.75	8m*8m
18	4080182	Bay of Bengal	8364*8062	2019/06/28	QPSI	26.21∼28.93	8m*8m
19	4062567	Arabian Gulf	6201*8062	2019/06/17	QPSI	35.44∼37.26	8m*8m
20	3489309	the Yellow Sea	8068*7882	2018/09/30	QPSI	31.03∼33.18	8m*8m
21	3266328	Gulf of oman	4719*5942	2018/06/18	QPSI	38.52∼39.71	8m*8m
22	3769375	the South China Sea	4709*6014	2019/01/17	QPSI	38.79∼ 39.99	8m*8m
23	6125890	the Yellow Sea	4703*6014	2021/12/10	QPSI	38.62∼39.82	8m*8m
24	6066586	Bohai Sea	5705*4288	2021/11/21	QPSI	48.18∼49.31	8m*8m
25	6204398	Bohai Gulf	5934*6858	2022/01/08	QPSI	36.75∼38.23	8m*8m
26	3489267	Hanzhou Bay	7594*7370	2018/09/30	QPSI	30.89∼ 32.92	8m*8m
27	5663965	Black Mud Bay	5852*7373	2021/05/24	QPSI	44.82∼ 45.94	8m*8m

These data were selected because they covered a wide range of incidence angles and included different marine environments.

The second part verified the superiority of SCANet. First, we present the dataset utilized in this section and the relevant experimental configurations. Next, the measures used to evaluate the performance of the algorithm are listed. Subsequently, the proposed SCAM is tested using the GF-3 dataset. Finally, SCANet and other target detection networks were compared using the GF-3 and SSDD datasets to verify their generalizability and effectiveness. We plot the detection results on an intensity image to obtain a good visual effect.

B. Comparison of Four-Component Decomposition Models

1) Qualitative Analysis: An important point for detection is whether there is a clear difference between the sea and the ship. Therefore, in this section, we use the distribution of the target and clutter pixels in 3-D space to evaluate the ship-sea differences between the six decomposition models. It should be noted that to maximize the separation of ship and clutter pixels, we manually segmented the scene and plotted the obtained components corresponding to the ship and clutter pixels as a 3-D scatter plot. Before drawing, we obtained the square root of each component to compress its dynamic range and obtain a better visual representation. As examples, we randomly selected two complex scenarios with multiple ships, as shown in Figs. 8 and 9. The Vol and Hlx power values of G4U were both approximately 0.01, making the ship pixels closer to the Dbl side of the coordinate axis, as shown in Fig. 8(c); an intersection exists between the clutter and ship pixels. Moreover, in Y4R and S4R, the Dbl power values of the ship pixels were also negligible and were mainly distributed near the coordinate origin, making the ship pixels closer to the Vol axis. Similarly, it can be seen in Fig. 8(f) and (g) that there is a cross section between the sea clutter and ship pixels in the Vol component. Fig. 8(b) and (d) shows that the distributions of both the BF4 and Y4O components are similar, and there is a clearer separation boundary between sea clutter and ship representations; however, the Dbl and Vol components gather many ship pixels in the interval [0, 0.2]. In Ship-4SD, the boundaries of sea clutter and ship were clearer than those in BF4 and Y4O, whereas the Dbl and Od components gathered fewer pixels in the interval [0, 0.2], as shown in Fig. 8(h). A similar situation occurred in the test scenario, as shown in Fig. 9. The test results for G4U are consistent with those in Fig. 8. For BF4, Y4O, Y4R, and S4R, an obvious intersection between ship and clutter pixels leads to an inability to effectively distinguish ship pixels from clutter pixels. In contrast, there is a clear boundary between the ship pixels and clutter pixels in Ship-4SD.

An aggregation of the above experimental results and analyses shows that the Od and Asym components in Ship-4SD describe the scattering characteristics of the ship more reasonably. We further analyzed the reasons for this. On the one hand, as analyzed earlier, the Od component in Ship-4SD characterizes the composite scattering structure consisting of even-bounce and odd-bounce reflectors on the ship; on the other hand, the Asym component makes complete use of the four quantities related to asymmetric scattering characteristics in the coherence matrix, which enables a more complete description of the asymmetric scattering characteristics of the ship and thus effectively compensates for the limitations of the Hlx component. It is well known that the core of ship detection lies in the separation of the ship and sea clutter based on the difference in the scattering characteristics between the ship and sea. In summary, compared with the other five decomposition

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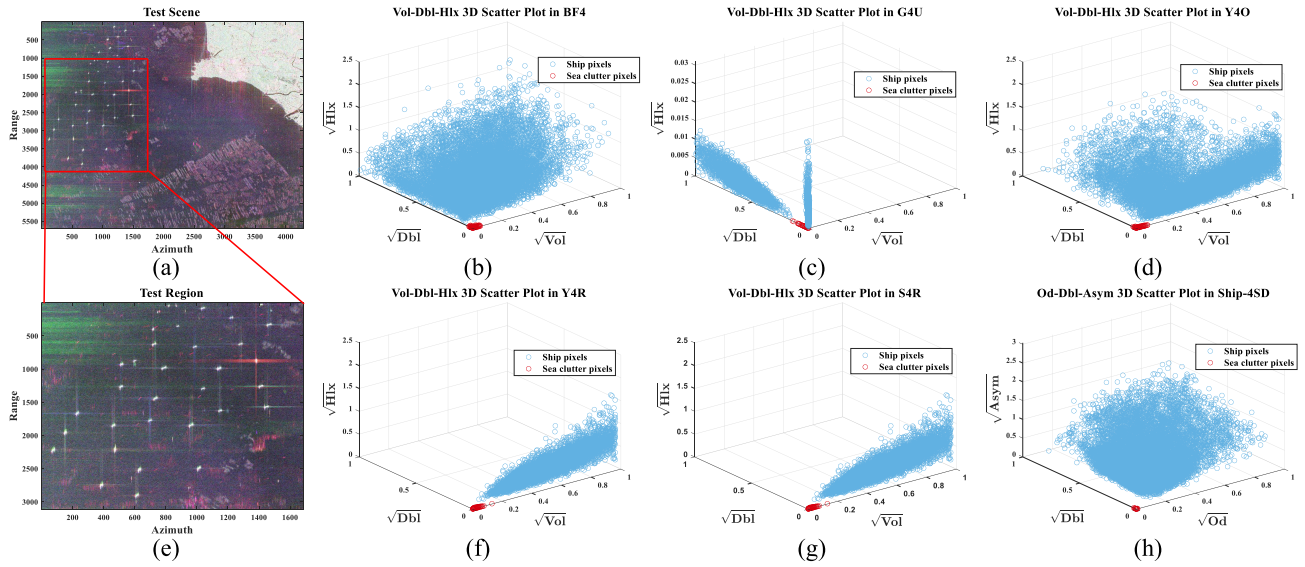


Fig. 8. Three-dimensional scatter plots of the ship and clutter pixels Dbl, Vol/Od, Hlx/Asym components in the test scene No. 2 (Product ID: 4128590). (a) Test scene, (b) BF4, (c) G4U, (d) Y4O, (e) test region, (f) Y4R, (g) S4R, and (h) Ship-4SD. The color of (a) and (e) is coded by $|S_{HH} - S_{VV}|_{\text{red}}$, $|S_{HH} + S_{VV}|_{\text{blue}}$, and $2|S_{HV}|_{\text{green}}$.

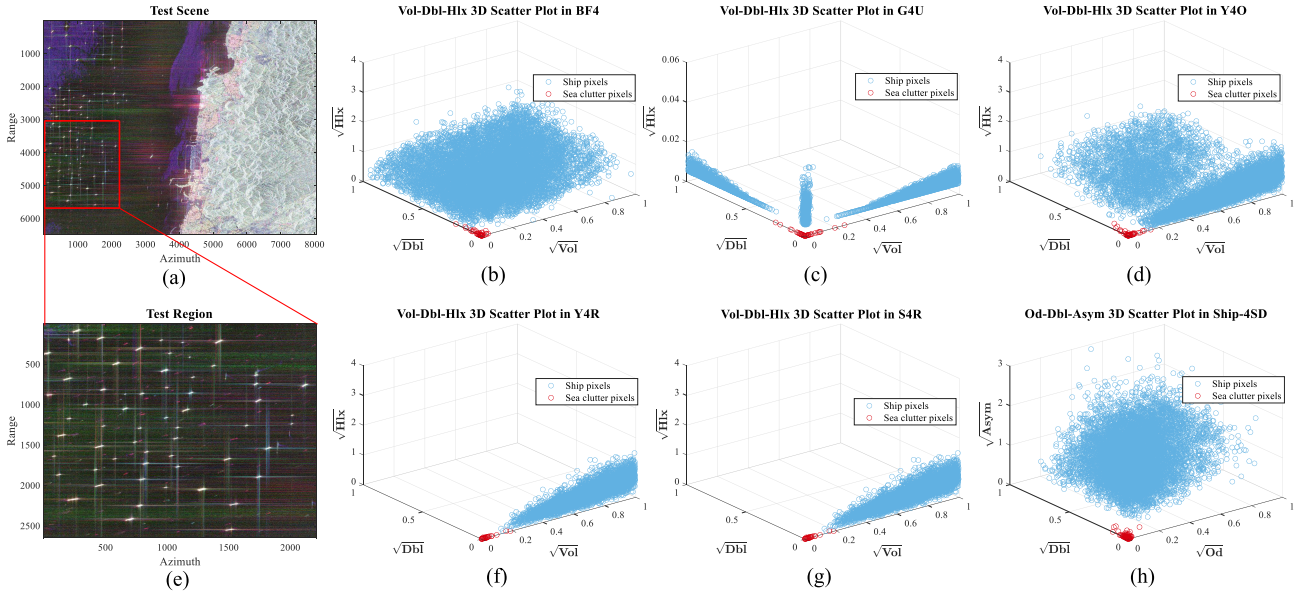


Fig. 9. Three-dimensional scatter plots of ship and sea clutter pixels of Dbl, Vol/Od, Hlx/Asym components in the test scene No. 3 (Product ID: 4056567). (a) Test scene, (b) BF4, (c) G4U, (d) Y4O, (e) test region, (f) Y4R, (g) S4R, and (h) Ship-4SD. The color of (a) and (e) is coded by $|S_{HH} - S_{VV}|_{\text{red}}$, $|S_{HH} + S_{VV}|_{\text{blue}}$, and $2|S_{HV}|_{\text{green}}$.

models, the proposed Ship-4SD decomposition algorithm has a more perceptible ship-sea boundary in the main scattering mechanisms, Dbl, Od, and Asym, as presented for the test ship, making it more suitable for application in the detection of ships at sea (Figs. 8 and 9).

2) Quantitative Evaluation: The main goal of this section is to evaluate the contribution of the four components to the ship's scattering characteristics representation and validate the rationality of the treatment of the negative power problem. To ensure the accuracy of the assessment, the ship was filtered out by manual segmentation, and a statistical analysis of each scattering component of the ship was performed.

a) Quantitative evaluation of contribution rate: Likewise, several representative examples have been randomly given. Fig. 10 shows the decomposition results for the six decomposition models of a representative example. Fig. 11 further shows the statistical results for the four components in patch A. It can be observed that P_h accounts for only 1% in BF4, G4U, Y4O, Y4R, and S4R, whereas the power of the Asym component describing nonreflective symmetric scattering in Ship-4SD accounts for 10%, which is significantly higher than that of the Hlx component. This is reasonable because a ship is composed of metallic materials, such as steel, with complex scattering characteristics. Hlx alone cannot completely describe the asymmetric scattering characteristics of a ship; hence, it contributes less to the representation of the ship. In addition, the percentages of the power P_v of the Vol component in BF4, G4U, Y4O, Y4R, and S4R were 2%, 3%,

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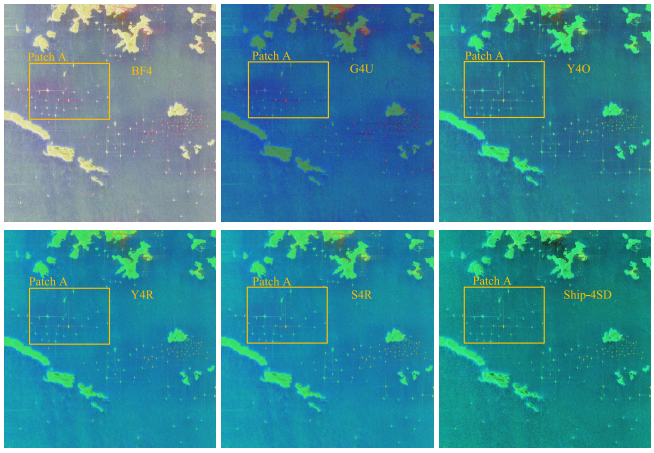


Fig. 10. Decomposition results of six decomposition models in scene No. 4 (Product ID: 3564287). The RGB color-codes: blue: Odd, green: Vol or Od, and red: Dbl.

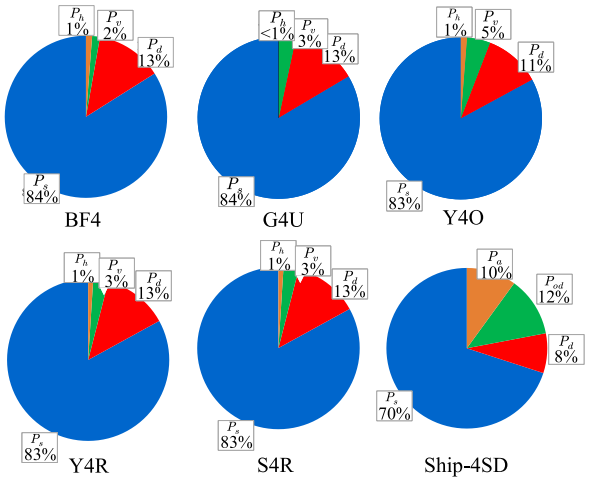


Fig. 11. Scene six statistical results for the four components of the six decomposition models in patch A.

5%, 3%, and 3%, respectively, in terms of their contributions to the ship's representation. The ratio of the Od component's power P_{od} in Ship-4SD was 12%, which was significantly higher than the ratio of Vol's power. This is reasonable because the Vol component describes a random dipole cloud structure similar to a tree canopy or grass, which rarely exists on ships. Ships are often composite scatters consisting of a deck (odd-bounce reflector) and linear steel objects (45°-oriented dipole). In summary, the experimental results presented in Figs. 10 and 11 confirm that Od and Asym in Ship-4SD are more suitable for describing ship scattering characteristics than other four-component models.

Another representative example is presented in Fig. 12, which shows the decomposition results for six decomposition models. The statistical results for Patch A are shown in Fig. 13, which shows the power shares of the Hlx component in BF4, G4U, Y4O, Y4R, and S4R was 6%, <1%, 6%, 4%, and 4%, respectively. The power ratio of the Asym component in Ship-4SD was 27%, which was considerably larger than the power shares of the Hlx component in the other five decomposition

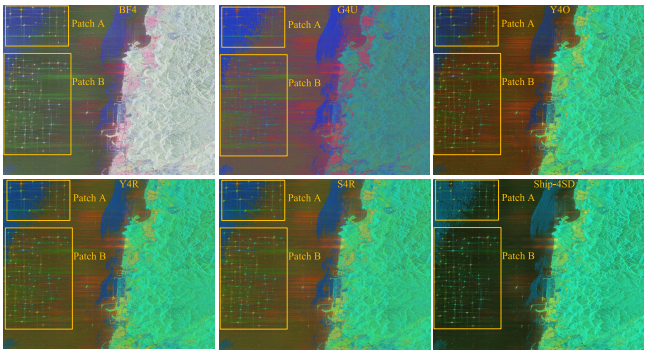


Fig. 12. Decomposition results of six decomposition models in scene No. 3 (Product ID: 4056567). The RGB color-codes: blue: Odd, green: Vol or Od, and red: Dbl.

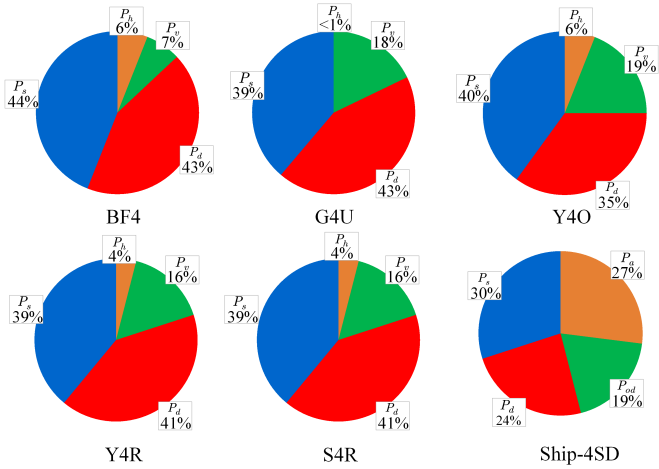


Fig. 13. Scene eight statistical results for the four components of the six decomposition models in patch A.

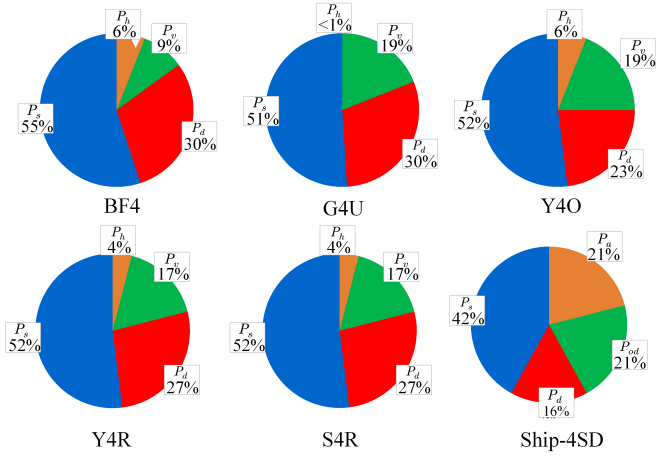


Fig. 14. Scene eight statistical results for the four components of the six decomposition models in patch B.

models. The power of the Od component in Ship-4SD was also 12%, 1%, 3%, and 3% higher than that in BF4, G4U, Y4R, and S4R, respectively. This is consistent with the percentage of the Vol component in Y4O (19%). Similar statistical results were observed for patch B (Fig. 14), where the power share of the Asym component in Ship-4SD was substantially higher than that of the helix component's scattering power share

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TABLE II

PIXELS WITH ODD AND DBL COMPONENT POWER GREATER THAN 0 IN THE PURE SEA AREA AND THE POWER RATIO OF ODD AND DBL

Patch	Product ID	r_s (%)	r_d (%)	r_{P_s} (%)	r_{P_d} (%)
1	6154474	100	3.32	94.67	0.02
2	6154462	100	3.88	93.79	0.03
3	2599253	100	3.81	93.42	0.03
4	6154442	100	3.71	93.54	0.03
5	6154473	100	4.5	92.50	0.04
6	4128588	100	3.52	90.97	0.02
7	4128582	98.98	3.97	90.61	0.03
8	6191496	89.4	62.38	44.75	15.48
9	4136813	85.7	62.5	65	5.83
10	4138161	86.75	61.89	59.33	6.79
Average of 100 scenes		97.08	22.7	71.72	2.57

* r_s and r_d denote pixels with a power greater than 0 in the Odd and Dbl components, respectively, and r_{P_s} and r_{P_d} denote the power shares in the Odd and Dbl components, respectively.

TABLE III

PERCENTAGE OF FOUR COMPONENTS POWER IN 55 SHIP-DENSE REGIONS AND PROPORTION OF ODD AND DBL POWER PIXELS GREATER THAN 0

Patch	Product ID	r_s(%)	r_d(%)	r_{P_s}(%)	r_{P_d}(%)	$r_{P_{od}}$(%)	r_{P_a}(%)
1	6154462	59.21	84.66	7.71	42.69	9.16	40.44
2	6191496	44.57	92.13	4.76	55.21	5.95	34.08
3	4080182	63.81	79.37	20.82	32.66	13.75	32.78
4	4062567	58.29	87.3	11.03	39.77	17.09	32.11
5	3489309	46.33	81.63	12.5	42.6	6.98	37.92
6	3266328	48.36	87.18	6.55	72.79	2.93	17.73
7	3769375	70.47	61.16	22.94	42.76	13.99	20.31
8	6125890	53.48	81.38	8.68	46.75	5.68	38.9
9	6066586	66.47	76.39	22.37	45.9	10.06	21.68
10	6204398	57.4	82.1	19.58	48.04	8	24.38
Average of 55 scenes		67.3	66.62	36.75	22.27	15.42	26.99

* r_s and r_d denote pixels with a power greater than 0 in the Odd and Dbl components, respectively, and r_{P_s}, r_{P_d}, $r_{P_{od}}$, r_{P_a} denote the power share in the Odd and Dbl, Od and Asym components, respectively.

in the other five decomposition models. The Od component scattering in Ship-4SD was 21%, which was 12%, 2%, 2%, 4%, and 4% higher than those in BF4, G4U, Y4R, and S4R, respectively.

By aggregating the above two scenes and three patches, we found that the three main scattering mechanisms (Dbl, Od, and Asym) presented in the ship using Ship-4SD had significantly increased power percentages. The power percentages of the Od and Asym components were significantly higher than those of Vol and Hlx. As analyzed earlier, the scattering structure on a ship is a composite scattering structure consisting of upright linear steel objects, such as railings, masts (even-bounce reflectors), and decks (odd-bounce reflectors), which can be well described using the Od component. Therefore, the power ratio of the Od component of the ship was greater than that of the Vol component. In addition, the ship scattering structure is complex, and some basic scattering mechanisms other than helix scattering exhibit nonreflective symmetric scattering characteristics. The Asym component was used to describe all asymmetric scattering mechanisms on the ship, making its power contribution considerably larger than that of the Hlx component. Therefore, compared with the other five decomposition algorithms in the experiment, the proposed Ship-4SD decomposition algorithm can reasonably and comprehensively describe the scattering characteristics of a ship and is more suitable for applications on ships.

b) Validation of the rationality of the negative power problem treatment: When solving for P_s and P_d, their powers may be negative. When P_s or P_d are negative, we set $P_s = 0$ or $P_d = 0$ under the constraint of the total power P_t to ensure that it is constant. In this section, we verify the reasonableness of this negative-power treatment. As the true values of the Odd, Dbl, Od, and Asym components within each resolution cell were unknown, a statistical analysis approach was used to show the effect of this treatment on the model decomposition results.

First, we counted the power percentage of the Odd and Dbl components in 100 pure sea regions in 100 scenes, and due to the article page limit constraints, we only listed the specific values of the Odd and Dbl components in ten scenes and their average values in 100 scenes, as shown in Table II. From Table II, it can be seen that the average percentage of pixels with a power greater than zero in the Odd component of the pure sea region is 97.08%, whereas the average percentage of pixels with a power greater than zero in the Dbl component is 22.7%. Furthermore, the average power proportion of the Odd components over 100 scenes was 71.72%, whereas that of the Dbl component was only 2.57%. The percentage of pixels with power greater than zero and the percentage of power values were both significantly larger in the pure sea region for the Odd component than for the Dbl component. This phenomenon is reasonable because, in a pure sea environment, the surface scattering mechanism dominates, leading to the most significant power share corresponding to a few negative power pixels of only 2.92%, whereas, for the Dbl component, the double-bounce scattering mechanism is not the dominant scattering mechanism in the sea. Hence, the share of Dbl in both metrics is negligible. In addition, for the pure sea surface, we used the positive and negative relations of $\Re\langle S_{HH}S_{VV}^*\rangle$ to determine whether the primary scattering mechanism is surface scattering. First, we solve for the value of P_s. When the Dbl component P_d is negative in the later solution steps, the power value of the Odd component is $P_s = P_t - P_{\text{od}} - P_a$. Therefore, the negative power of P_d has minimal effect on P_s.

We also considered 55 ship-dense scenes with different satellite signal incidence angles, sea conditions, and ship directions. In these 55 scenes, the percentages of pixels with Odd and Dbl component power greater than 0 were 67.3% and 66.62%, and the power percentages were 36.75% and 22.27%, respectively, as shown in Table III. Compared to the pure sea surface, the percentage of pixels with power greater than zero and the percentage of power in the Odd component on the test ship decreased significantly, whereas the percentage of these two in Dbl increased noticeably. This is reasonable because of the complex structure of ships. Although surface scattering occurs on ships, it is no longer the primary scattering mechanism. Instead, the remaining three scattering mechanisms were the main mechanisms. This also shows that using the positive and negative relationships of $\Re\langle S_{HH}S_{VV}^*\rangle$ enables the correct determination of whether surface scattering or double-bounce scattering mechanism prevails within a resolution cell. In addition, our interest in ships is more in the power of the Dbl, Od, and Asym components because these are key to ship detection.

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Statistical analysis results showed that the Dbl, Od, and Asym components accounted for substantially more power than Odd. More importantly, when we use $\Re\langle S_{HH}S_{VV}^{*}\rangle$ to determine whether the double-bounce scattering mechanism dominates within a resolution cell, we first solve P_d. Therefore, even if P_s is negative in the latter step, it does not affect the previously solved P_d.

In summary, the foregoing statistical results show that the decomposition method used in this study to deal with the appearance of negative power in the Odd and Dbl conditions is reasonable. This treatment does not change the scattering mechanism inherent in a resolution cell and thus has a negligible effect on the decomposition results.

C. Comparison of Multicomponent Model Decomposition Results

In Section IV-B, the qualitative analysis and quantitative evaluation reveal that the Ship-4SD model is more suitable for describing the scattering characteristics of a ship, and the three related components, Dbl, Od, and Asym, contribute more to ship representation than the other four-component decomposition models.

However, it should be noted that the field of PolSAR target decomposition was developed for seven components. Intuitively, more components are expected to provide a finer description of the scattering structure of the ground, which is more beneficial to the interpretation of the ground. Multicomponent decomposition [16], [17] has been proven to be more suitable for describing vegetation and urban areas in experiments using ALOS-2/PALSAR-2 and TerraSAR-X data. However, these advanced models have not yet been applied to ships or discussed in the literature. In addition, it can be expected that with an increasing number of components, the scattering structure of a ship can be described more finely. However, whether this increased level of refinement leads to an improved detection performance remains unknown and has not been explored in previous studies. Furthermore, the efficiency of the model computation may decrease because of the increased number of model components. This section examines these concerns.

To provide a convincing comparison, we compared two current state-of-the-art (SOTA) decomposition models: 6SD [16] containing Odd, Dbl, Vol, Hlx, Od, and Cd (compound dipole scattering) components and 7SD [17] containing mixed dipole scattering (Md) in addition to 6SD. For a fair comparison, we treated (Vol + Cd + Hlx) in 6SD and (Vol + Cd + Md + Hlx) in 7SD as a "Group" and compared with Asym in Ship-4SD, while the remaining three components remained the same.

1) Ship-Sea Difference Analysis: The most important factor in ship detectability is whether there is a clear difference between ships and the sea. As direct and visual indicators, the distribution of ship and clutter pixels in 3-D space can be used to evaluate the ship-sea differences in the three decomposition models.

We tested 14 scenarios under various sea conditions, incidence angles, and ships. Due to space constraints, only two representative scenes were presented in this study

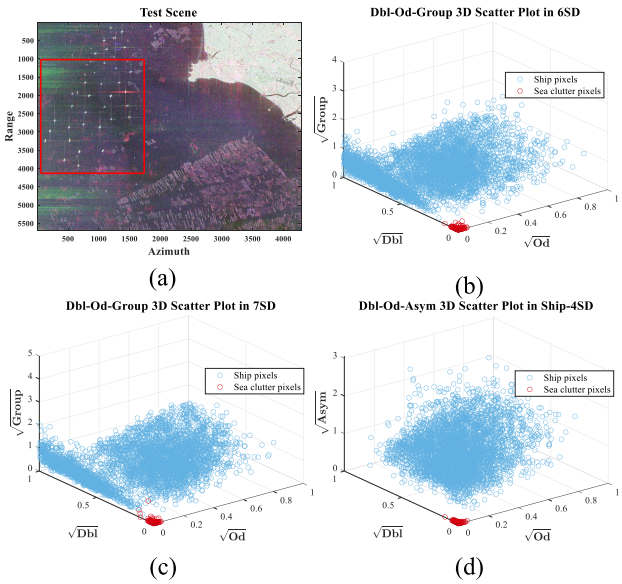


Fig. 15. Three-dimensional scatter plots of the three models for a representative example. (a) Test scene (Product ID: 4128590). (b) 6SD. (c) 7SD. (d) Ship-4SD. The color of (a) is coded by $|S_{HH}-S_{VV}|_{\text{red}}$, $|S_{HH}+S_{VV}|_{\text{blue}}$, and $2|S_{HV}|_{\text{green}}$.

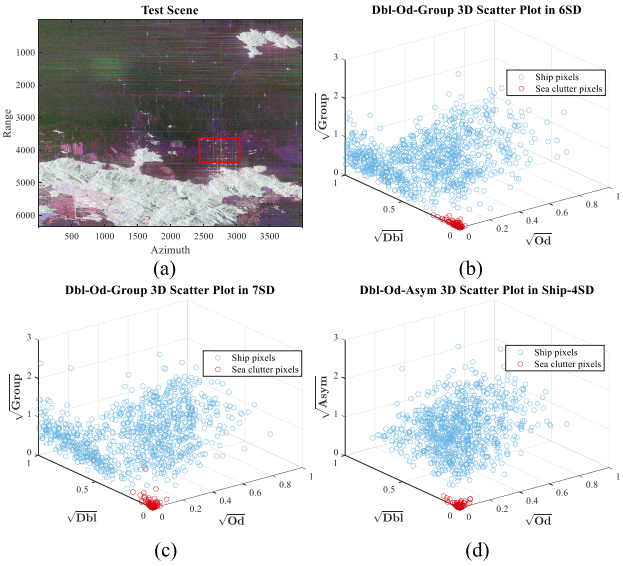


Fig. 16. Three-dimensional scatter plots of the three models for another representative example. (a) Test scene (Product ID: 5691180). (b) 6SD. (c) 7SD. (d) Ship-4SD. The color of (a) is coded by $|S_{HH}-S_{VV}|_{\text{red}}$, $|S_{HH}+S_{VV}|_{\text{blue}}$, and $2|S_{HV}|_{\text{green}}$.

(the remaining scenes had similar results). Similar to the previous experiment, the Odd component describes the scattering characteristics associated with the ocean surface; thus, it cannot be applied to ship detection [39], [40]. Therefore, it does not appear in the 3-D space. Figs. 15 and 16 show the 3-D scatter plots between a test ship and sea clutter in the three models. Visually, Ship-4SD showed more differences between the ships and clutter in both scenes; the distance between the ship and clutter pixels was greater. For the 6SD and 7SD models, as shown in Fig. 15(b) and (c) the distance between the ship and clutter pixels is small. Similarly, as shown in Fig. 16(b) and (c), there are intersections between the ship

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TABLE IV
AVERAGE TCR OF THE SHIP AND CLUTTER IN THE THREE COMPONENTS

Component	6SD (dB)	7SD (dB)	Ship-4SD (dB)
Dbl	32.41	32.99	**33.50**
Od	28.66	**29.05**	28.94
Group/Asym	29.53	29.92	**30.56**

TABLE V
AVERAGE ODD POWER VALUES FOR THE THREE MODELS IN 344 SHIPS

Component	6SD (dB)	7SD (dB)	Ship-4SD (dB)
Odd	63.86	62.56	**59.61**

and clutter pixels, which are mixed with each other. Hence it is not conducive to ship detection. For Ship-4SD, the boundary between the ship and the clutter was more apparent in both scenarios, indicating that the ship-sea differences were great.

The experimental results above intuitively show that the Ship-4SD model has greater ship-sea differences when a combined analysis of Dbl, Od, and Asym/Group is used to evaluate the ship-sea differences of the three models from a visual observation point of view. In contrast, the ship-sea differences between 6SD and 7SD were smaller, and there was an intersection between the ship and clutter pixels. In summary, the above experiments further verify that Ship-4SD is more suitable for describing the scattering characteristics of ships.

2) Target-Clutter-Ratio (TCR) Comparison: Based on the above experiments, we consider why the ship-sea differences in the three components related to ship scattering characteristics are larger in Ship-4SD. In SAR image target detection, TCR is the core metric used to evaluate the performance of the detector [41], and can be expressed as [42]

$$\mathrm{TCR} = 10\log_{10}\left(\frac{I_T}{I_C}\right)(\mathrm{dB}) \tag{23}$$

where I_T denotes the power value of the component in the ship and I_C is the power value of the component in the clutter.

The average TCR values for the three components (Dbl, Od, and Asym/Group) are listed in Table IV. We counted 344 ships of different types and headings under 14 scenarios. From Table IV, it can be observed that in the Dbl component, the TCR values one for 6SD and 7SD are 32.41 and 32.99 dB, respectively; while the average TCR value of the Dbl component in Ship-SD is 33.50 dB, which is 1.09 dB and 0.51 dB higher than that of 6SD and 7SD, respectively. In the Asym/Group component, the TCR values of the three models were found to be 29.53, 29.92, and 30.56 dB, respectively, with Asym having the highest TCR value; while in the Od component, 7SD had the highest TCR of 29.05 dB, followed by Ship-4SD with 28.94 dB, and the lowest is 6SD with 28.66 dB.

In addition, from the decomposition results of the three models we found the power differences of the three components, Odd, Od, and Dbl, in Ship-4SD, 6SD, and 7SD were large. More specifically, the Odd and Dbl powers in Ship-4SD were lower than 6SD and 7SD, while the Od power was greater than 6SD and 7SD. The reasons for this are twofold. 1) From the perspective of the solution process of the decomposition algorithm, the power values of the four components Hlx, Od, Cd, and Md in 6SD and 7SD are first obtained directly from the elements of the coherence matrix and optimized with respect to each other. Then, the power values of Vol, and finally, Dbl and Odd. Hlx, Od, Cd, Md, and Vol determine the power values of Dbl and Odd, respectively. In Ship-4SD, the power of Od is obtained directly from the elements of the coherence matrix but is not optimized, whereas the power of Asym is obtained from T_{33}, and the powers of Odd and Dbl are solved. Therefore, the difference between the Asym and the (Vol + Cd + Md + Hlx) power values led to a difference between the Odd and Dbl power values. 2) From the perspective of the description object of each component, the Vol, Cd, and Md components describe the symmetric scattering characteristics of the ship; the Hlx component describes the asymmetric scattering characteristics of the ship, and the Asym component describes all the asymmetric scattering characteristics of the ship. In Section II-A, we analyzed the inherent defects of Hlx, which cannot fully describe all asymmetric scattering characteristics of the ship. Therefore, the power of Hlx is not equal to that of Asym, and the power of (Vol + Cd + Hlx) or (Vol + Cd + Md + Hlx) is not equal to that of Asym because of the different components that describe the different structures. (Vol + Cd + Hlx) or (Vol + Cd + Md + Hlx) and the Asym power are not equal, which also leads to a difference between the Odd and Dbl power values.

From the scenarios we tested, we can confirm that in both the Dbl Asym/Group components, Ship-4SD has higher average TCR values than 6SD and 7SD, whereas in the Od component, Ship-4SD is slightly lower than 7SD and higher than 6SD. Note that TCR values of the Od component between Ship-4SD and 7SD have a negligible difference, as shown in Table IV. Thus, in general, the ship-sea difference is greater for the Ship-4SD model component. Next, we analyze the experimental phenomena in terms of individual components.

1) Odd component: Among the 344 ships tested, the Odd component's power values of the Ship-4SD model decreased relative to those of the 6SD and 7SD models, as listed in Table V. The component power values associated with the surface scattering characteristics were more suppressed in Ship-4SD, whereas the scattering component power values associated with the ship scattering characteristics were enhanced. As mentioned earlier, the complex scattering structure of the ship produces different types of scattering mechanisms; therefore, more Odd power values are assigned to other components in Ship-4SD, resulting in a decrease in the power of the Odd component associated with the sea surface scattering characteristics. This indicates that the Ship-4SD model is more appropriate for describing ships.
2) Dbl component: In the Dbl, the average TCR values of Ship-4SD are higher than those of 6SD and 7SD, which

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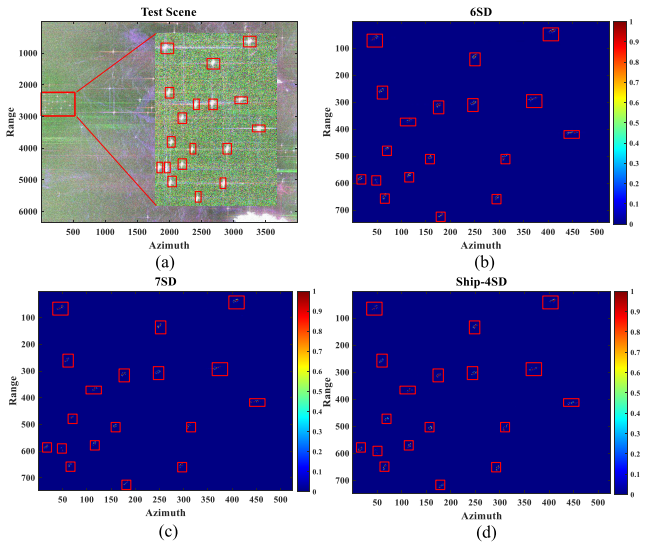


Fig. 17. Results of describing the scattering characteristics of ships by the Dbl components in the three models. (a) Test scene (Product ID: 5691179). (b) 6SD. (c) 7SD. (d) Ship-4SD.

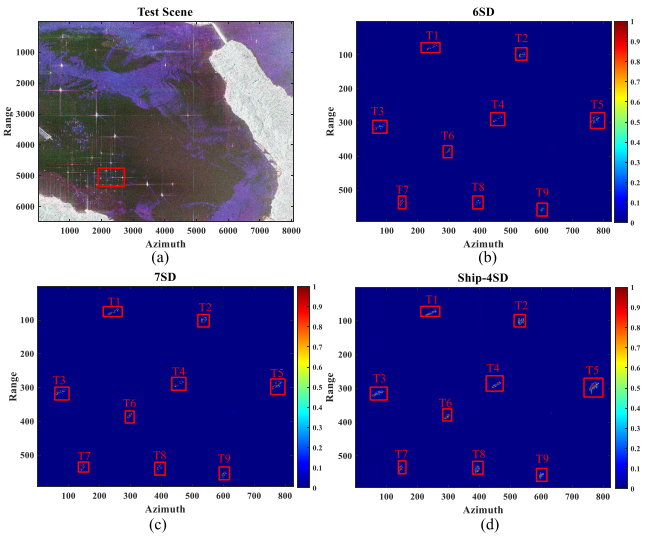


Fig. 18. Results of describing the scattering characteristics of ships by the Od components in the three models. (a) Test scene (Product ID: 4008613). (b) 6SD. (c) 7SD. (d) Ship-4SD.

means that the ship-sea differences in this component are also greater in Ship-4SD. However, it should be noted that the ratio of the Dbl component to the total power in Ship-4SD decreases, but at the same time, the power share of the Dbl component in the sea clutter is reduced. As mentioned previously, the Dbl component is closely related to the scattering characteristics of the ship and cannot effectively describe the scattering characteristics of the sea surface; therefore, its power value in sea clutter is reduced in proportion. It can also be seen from Fig. 17 that the Dbl component of all three models effectively describes the scattering characteristics of the ship and that the ship can be identified by relying on the Dbl component alone, indicating that this component is a useful detection quantity for ship detection.

3) Od component: In the Od component, the average TCR value of 7SD was the highest for the three decomposition models, followed by Ship-4SD, while that of 6SD was the lowest. The reason for this difference was that although the power values of the Od component were obtained from the real part of T_{13} in Ship-4SD, 6SD, and 7SD, the 7SD also uses the Hlx, Cd, and Md components to optimize Od additionally and adequately, whereas the absence of these components in Ship-4SD makes similar optimization impossible, resulting in a lower average TCR for the Od component in Ship-4SD than in 7SD. In 6SD, although the same optimization was performed using the Hlx and Cd components, the optimization was incomplete because of the lack of Md components in 6SD, resulting in the lowest TCR. However, as shown in Table IV, this difference is small, and there is almost no difference between Ship-4SD and the SOTA seven component approach using the Od component, considering the actual solution factors. In addition, we can see from Fig. 18 that the ship can be identified using only the Od component. A closer look at the ships in Fig. 18 shows that the shape contours of the ships are more complete and the pixels are brighter in Ship-4SD, particularly in the T4, T5, and T6 targets, indicating that Od is also a useful detection quantity.

4) Asym/Group component: In the Asym/Group component, the mean TCR of Asym was higher than that of "Group" in both 6SD and 7SD as the "Group" is (Vol + Cd + Hlx) in 6SD and (Vol + Cd + Hlx + Md) in 7SD. From these components, Vol describes a random dipole cloud, which we analyzed earlier because of its inability to effectively describe the scattering characteristics of the ship, and Hlx, which we similarly analyzed in Section IV-C1 for its inherent defects. Therefore, the power values for these two components are small and do not reasonably characterize the ship. On the other hand, the power values of components Cd and Md originate from the imaginary part of T_{13} and the real part of T_{23}, respectively. Under complex sea conditions, these two components may be related to the scattering characteristics of the sea surface, which increases the power share of these two components of sea clutter. These two aspects lead to a reduction in the difference in "Group" values between the ship and the ocean, which results in a lower average TCR. The Asym component describes the asymmetric scattering characteristics. It is well known that asymmetric scattering is caused by the coherence of multiple scattered mechanisms. The ship scattering structure is complex, and the Asym component can accurately describe the asymmetric scattering characteristics of a ship. In contrast, more surface scattering mechanisms are produced on the sea surface, and Asym cannot describe this scattering property. Therefore, the average TCR value is high in the Asym component, and the ship-sea difference is large. As shown in Fig. 19, the nine ships in the scene can be described reasonably well by these three components correctly, allowing them to be

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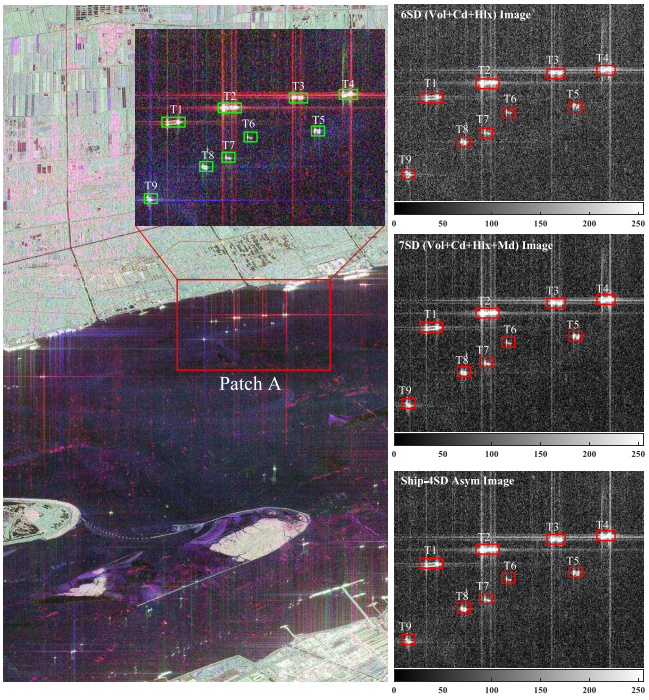


Fig. 19. Power plots for the three components of 6SD, 7SD, and Ship-4SD in the test scene No. 1 (Product ID: 5663989), with green boxes indicating ground truth and red boxes denoting the ships correctly characterized by Asym/Group. The color of the ground-truth image is coded by $|S_{HH}-S_{VV}|_{\text{red}}$, $|S_{HH}+S_{VV}|_{\text{blue}}$, and $2|S_{HV}|_{\text{green}}$. The values of the grayscale map are normalized between 0 and 255.

clearly presented on the power diagram. We also observe that the sea clutter scattering intensity is similar for the three decomposition models in the scene. As illustrated in Fig. 19 and listed in Table IV, the Asym is a useful component for ship detection.

In summary, we concluded that Ship-4SD is more suitable for describing ship scattering characteristics from the perspective of ship detection, and its suppression of clutter is more effective because the power of the surface scattering component is transferred to other components during decomposition. Second, in terms of target detection, the three components Dbl, Od, and Asym/Group were all valid detection quantities, and the three decomposition methods had comparable effects on Od. However, the three components in this study may be more favorable for detection, as shown in the analysis in this section and the scatterplots in Figs. 15 and 16 in Section II-A. Third, the above experimental results imply that advanced decomposition in target detection may not necessarily lead to an improvement in detection performance and that the four-component decomposition in this study may be relatively sufficient. However, with greater decomposition and finer target characterization, it is expected to be more useful for target recognition.

3) Computational Efficiency Comparison: In this section, we examine the computational efficiency of Ship-4SD, 6SD, and 7SD. Intuitively, 6SD and 7SD have more components, their algorithms contain more constraints, and their computational efficiencies are expected to be lower. For a fair comparison, we selected ten scenarios for testing. To avoid

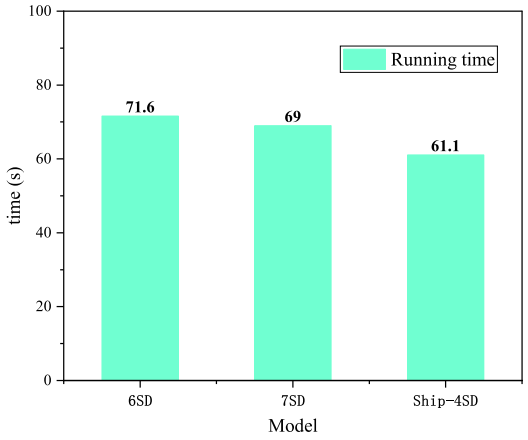


Fig. 20. Average time required to decompose a scene by three decomposition models.

errors caused by the instability of the test environment, these ten scenes were decomposed five times to obtain a decomposition time of five times, and the average time of each scene decomposed by the three algorithms was obtained, as shown in Fig. 20. As seen in the graph, Ship-4SD took the least time at 61.1 s, followed by 7SD at 69 s, and the most time was spent by 6SD at 71.6 s. Ship-4SD contains only four components with fewer judgment conditions and thus runs faster, while in 7SD there are more components and more judgment conditions, making the model decomposition slower; compared to 7SD, 6SD contains a rotation operation on the coherent difference matrix, thus making the computational efficiency of 6SD model lower than 7SD. Please note that the testing environment used in this study was an Intel[1] Core[2] i7-9700K CPU at 3.60 GHz, Windows 10, Microsoft Visual Studio Community 2017.

The experimental results of the 6SD, 7SD, and Ship-4SD decomposition models on the GF-3 dataset were combined, and the difference between ship and sea clutter was greater in the Ship-4SD model. The average TCR of the Ship-4SD components was relatively higher, and the model decomposition efficiency was faster in Ship-4SD. Therefore, the Ship-4SD decomposition model is superior to the multicomponent model considering both the model decomposition efficiency and core ship-sea differences in target detection.

D. SCANet Detection Performance Validation

1) Dataset and Settings: In this study, we produced a training set using 73 scenes of GF-3 fully polarized data with 2494 512 × 512 slices containing 7904 targets; the test set contained 46 scenes of GF-3 fully polarized data with a total of 427 512 × 512 slices containing 1424 ship targets. Multiscale ships with different incidence angles, sea conditions, and complex scenes were included in the GF-3 dataset. Ablation experiments were conducted on the GF-3 dataset to evaluate the effectiveness of SCANet. We validated the effectiveness and generalization of SCANet on a publicly available SSDD dataset. Multiple SAR ships exist with various

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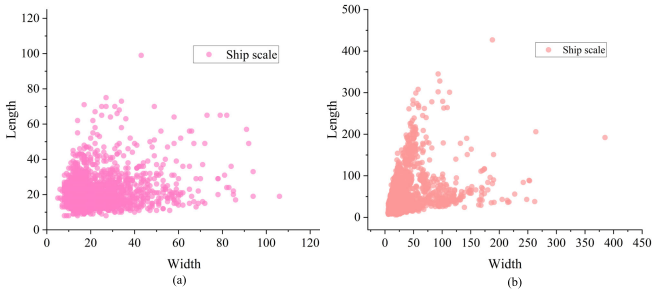


Fig. 21. Samples and scales in the scene (unit: pixel). (a) GF-3 dataset. (b) SSDD dataset.

polarization modes, image resolutions, sea states, sensors, and scenarios in the SSDD dataset [19]. It has a total of 1160 images, of which the training set contains 928 images, and the remaining 232 images are part of the test set. In the experiments conducted in this study, we resized all images in the SSDD dataset to 512 × 512 pixels. All label formats in the dataset follow the VOC2007 format. Fig. 21 shows the target scales in the two datasets. The target scale in the GF-3 dataset was small, whereas the ship scale in the SSDD dataset was larger, and the difference in scale between the ships was also larger. To accommodate the inputs to the network, we converted the results of the polarization decomposition into an image, and both of the above datasets were fed into the network as images.

In the SCAM, the feature maps of the input EPSAB are grouped four times; the number of channels in each group is 512, 128, 64, and 32; the corresponding convolution kernel sizes in each group are 3, 5, 7, and 9, respectively; and the dilation rate in the RFB is set to 1, 3, 5, and 7. The anchor box sizes of the GF-3 and SSDD datasets were set to 12, 28, 44, 64, and 128, and the anchor frame aspect ratios were set to 0.5, 1, and 2, respectively. In SCANet, the backbone networks of the model use ResNet50 with the corresponding pretraining weights, an Adam optimizer with an initial learning rate of 1e-4, an exponential decay strategy with the threshold of NMS set to 0.3 and a drop of 0.94 per epoch, stopping training early when the loss does not drop in more than 20 epochs, with a maximum epoch of 200, and the weights are saved every ten epochs. All experiments were performed using the TensorFlow framework on a computer with an Intel i7-9700k processor and a GeForce GTX 2070 GPU graphics card. The operating systems used were Windows 10, CUDA10.0, and cuDNN 7.6.4.

2) Evaluation Indicators: To quantitatively assess the performance of the proposed network, the Precision, Recall, average precision (AP), and $F1$ scores were utilized as evaluation criteria [20], [21], [23], [26]. The intersection-over-union (IoU) threshold and classification score were set to 0.5, that is, if the IoU between the prediction box and ground truth was greater than 0.5, and the classification score threshold was greater than 0.5, the target detected by the bounding box was the correct ship.

3) Ablation Study: Two main structures, the EPSAB and RFB, were included in the SCAM. Therefore, it is necessary to explore the performance of different structures to illustrate

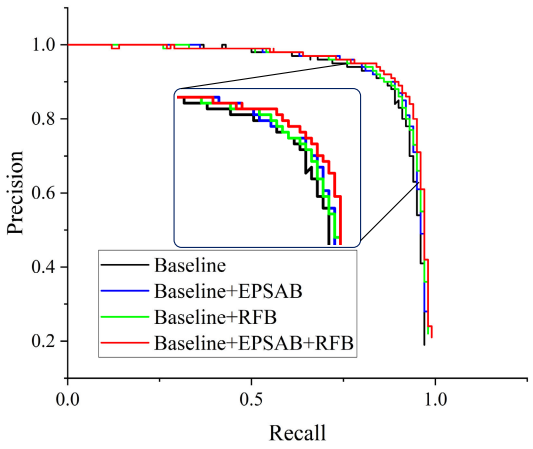


Fig. 22. PR curves corresponding to the four structures.

TABLE VI
IMPACT OF DIFFERENT COMPONENTS IN SCAM ON THE GF-3 DATASET

EPSAB	RFB	*Precision* (%)	*Recall* (%)	*F1*	*AP* (%)
×	×	93.00	81.40	0.87	92.69
✓	×	93.21	81.25	0.87	93.56
×	✓	94	82.73	0.88	93.66
✓	✓	**94.57**	**83.44**	**0.89**	**94.43**

their effectiveness. Under the same experimental environment and parameter settings, we trained and tested networks containing different structures and obtained the following results according to the evaluation metrics in Section IV-D2.

From Table VI, we can observe that the detection performance of any single structure improved with respect to the baseline. When we combined the EPSAB and RFB structures to form the SCAM (EPSAB + RFB), the detection performance was significantly improved. Precision, Recall, *F1* Score, and AP in SCAM were 1.57%, 2.04%, 0.02%, and 1.74%, higher than baseline, respectively. In addition, we plotted the detection results for the four structures in Table VI as Precision-Recall curves (PR curves), as shown in Fig. 22. The baseline is distributed in the innermost part, the SCANet containing the SCAM structure is distributed in the outermost part, and the curves containing the EPSAB and RFB structures are in the middle. Therefore, the PR curves also reflect that the network containing the SCAM structure exhibits the best detection performance.

The detection of small targets in complex scenes is a major challenge in neural networks [43], and the same is true in the field of target detection in SAR images. Ship detection in SAR images often relies on bright spots presented by the target; however, because of the small scale of ships in SAR images, less information can be provided. More seriously, when the scene includes land, objects on land that are similar in shape to the ship also affect the network detection performance. Therefore, to examine the detection performance of the proposed SCANet, we selected a scene with a complex background and a small target scale as an example. The detection results for these four structures were plotted on an intensity map, as shown in Fig. 23. In the first scene, which contains nine targets and part of the land background, Baseline generates a

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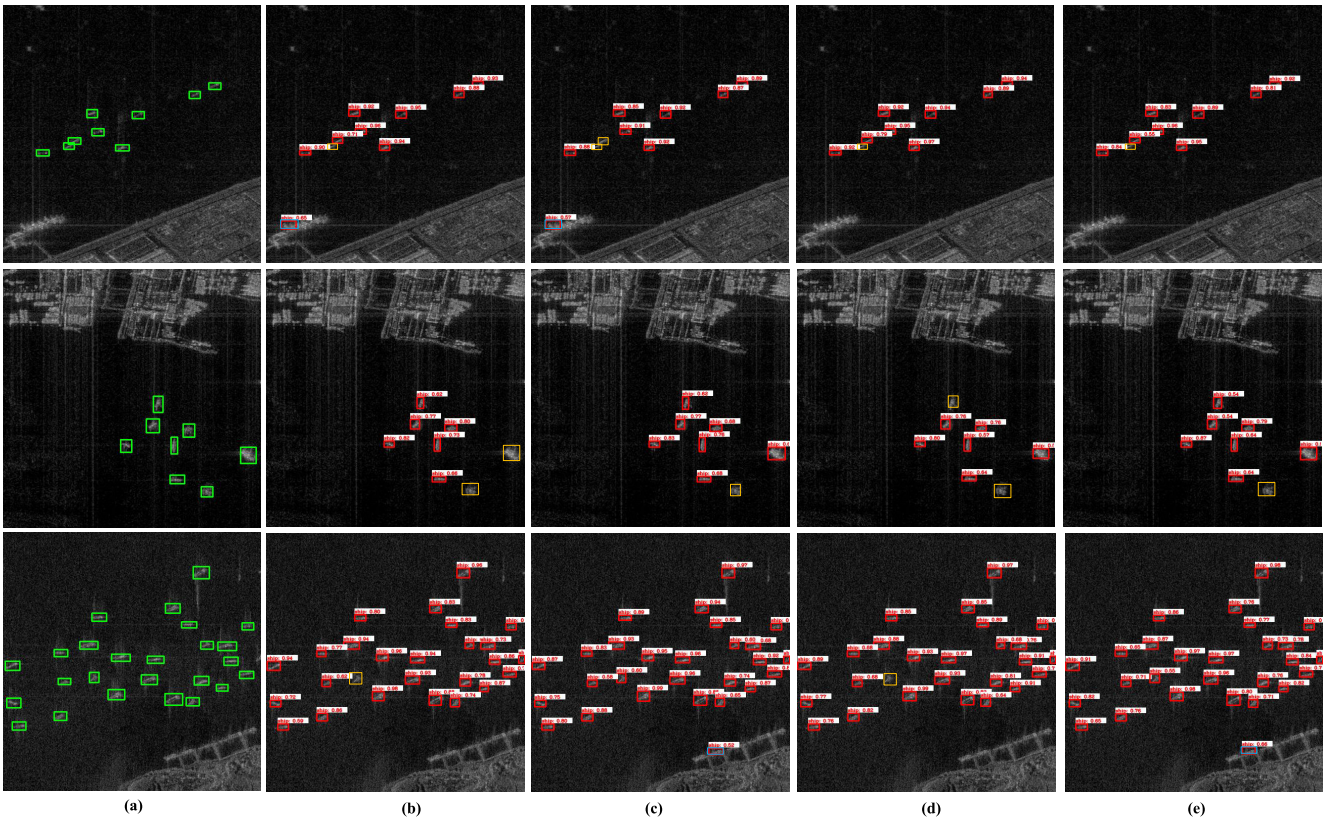


Fig. 23. Detection results of the four structures. The green box indicates the ground truth, the red box represents the predicted result, the yellow box refers to the missed detection target, and the blue box indicates the false alarm. The corresponding Product IDs of the three scenarios are 5663989, 5663965, and 3489267. (a) Ground-truth. (b) Baseline. (c) Baseline + EPSAB. (d) Baseline + RFB. (e) Baseline + EPSAB + RFB.

missed target and a false alarm; EPSAB misses two targets and generates a false alarm; and RFB and EPSAB + RFB miss one target. In the second scene, which contains eight targets and a land background, the Baseline and RFB miss two targets, whereas EPSAB and EPSAB + RFB miss only one. In the third scene, which contains many dense small-scale ships, Baseline and RFB each missed one target; EPSAB and EPSAB + RFB detected all ships but generated a false alarm.

Combining Table VI and Figs. 22, and 23, we can conclude that the SCAM (EPSAB + RFB) has the best detection performance and the highest accuracy in this article. As analyzed earlier, there are large differences in the shape, scale presented by the ship, and ship-sea contrast under different scattering mechanisms. Therefore, we used EPSAB in the SCAM to mine the polarization features of the ship under different scattering mechanisms and fused the captured features. In addition, to adapt to the multiscale variation of targets in the dataset, the RFB is added, which can increase the receptive field of the convolutional layer without changing the size of the convolutional kernel and thus capture local and global features on the feature map. Therefore, the SCAM can also perform excellently in complex scenarios involving small-scale ships.

4) Comparison of CNN-Based Target Detection Networks on GF-3 Dataset: We evaluated the overall performance of SCANet and compared it with those of other CNN-based target-detection networks to illustrate the effectiveness of the network. The networks compared were CenterNet [44], SSD-512 [37], YOLOv4 [45], YOLOv4 + SE, and LSDNet [26]. These target detection networks were chosen because they are representative of anchor-based and anchor-free networks in the detection of targets and have SOTA detection performance; and they are also widely used for comparison in the field of SAR target detection [31], [32]. LSDNet is a two-stage target-detection network driven by polarization characteristics. These six networks were quantitatively evaluated using the evaluation metrics described in Section IV-D2. The code of the network selected for the experiments is open source, and interested readers can refer to these documents [37], [44], [45].

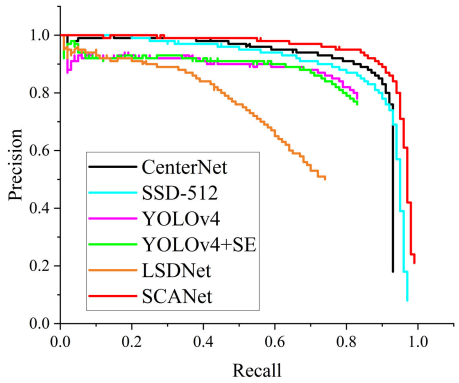


Fig. 24. PR curves of six networks on GF-3 dataset.

As shown in Table VII, the highest Precision of SCANet is 94.57%, which is 1.73%, 0.53%, 7.85%, 9.02%, and

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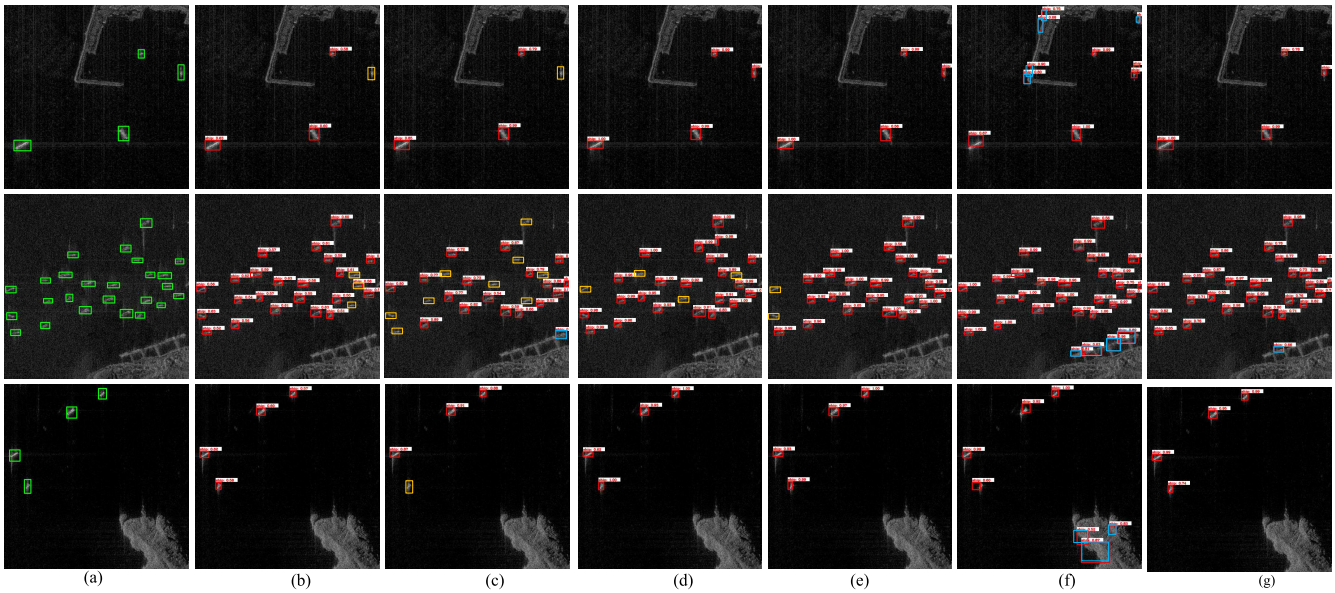


Fig. 25. Six network detection results. Green boxes indicate ground truth; red boxes indicate predicted results; yellow boxes indicate missed detection; and blue boxes indicate false alarms. The Product IDs of the four scenarios are 5663965, 3489267, and 3564287 in order. (a) Ground-truth. (b) CenterNet. (c) SSD-512. (d) YOLOv4. (e) YOLOv4 + SE. (f) LSDNet. (g) SCANet.

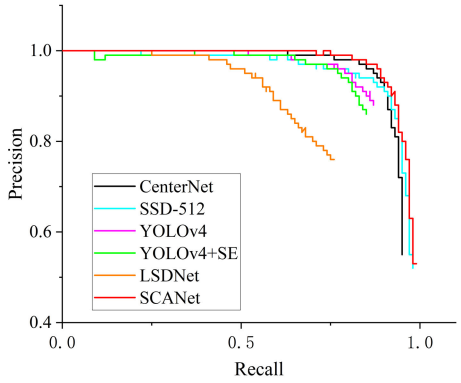


Fig. 26. PR curves of six networks on SSDD dataset.

TABLE VII

COMPARISON OF DETECTION RESULTS OF CNN-BASED TARGET DETECTION NETWORKS

Method	*Precision* (%)	*Recall* (%)	*F1*	*AP* (%)
CenterNet	92.84	74.98	0.83	89.18
SSD-512	94.04	58.91	0.72	89.45
YOLOv4	86.72	74.07	0.8	75.55
YOLOv4+SE	85.55	72.59	0.79	75.32
LSDNet	50.12	74.49	0.6	60.08
SCANet	**94.57**	**83.44**	**0.89**	**94.43**

TABLE VIII

DETECTION RESULTS OF SIX NETWORKS ON THE SSDD DATASET

Method	*Precision* (%)	*Recall* (%)	*F1*	*AP* (%)
CenterNet	**98.75**	72.48	0.84	93.83
SSD-512	95.89	77.06	0.85	94.65
YOLOv4	95.35	79.08	0.86	86.15
YOLOv4+SE	95.05	77.43	0.85	83.48
LSDNet	75.92	75.78	0.76	71.91
SCANet	97.81	**81.83**	**0.89**	**96.56**

44.45% higher than those of CenterNet, SSD-512, YOLOv4, YOLOv4 + SE, and LSDNet, respectively; the highest Recall is 83.44% for SCANet, while all other networks have a recall rate of less than 80%. For the *F1* Score, SCANet achieves 0.89, which is much higher than the other five networks; the same occurs for the metric AP, which is 94.43% for SCANet, the highest among these six networks. To demonstrate the performance of these six networks more intuitively, we plotted the corresponding PR curves, as shown in Fig. 24. The red color indicates the PR curve of the SCANet. It can be seen that the red curve wraps the curves of the other five networks, which means that the area enclosed by the PR curve of SCANet is larger, and the overall detection performance is better.

Furthermore, we plot the detection results of the six networks on the SPAN images, as shown in Fig. 25. Fig. 25 contains four scenes, three of which contain a land background and one is an open sea. In Scene 1, four small targets were included, and the land was similar in shape to the ship in Scene 1. In these scenes, YOLOv4, YOLOv4 + SE, and SCANet detected all the ships, whereas CenterNet and SSD-512 missed one target. For LSDNet, although all ships were detected, the land was recognized as a ship, resulting in five false alarms. Similarly, in Scene 2, LSDNet detected a large number of false land alarms in scenes with a land background, resulting in dramatic degradation of its detection performance.

Through the previous analysis, the reason for the dramatic degradation of the LSDNet detection performance in complex scenarios can be explained. On the one hand, LSDNet, as a two-stage target detection network, incorrectly considers land as a candidate region when making region proposals in the first stage, leading the network to provide the possibility for these land features to become false alarms in the second stage of prediction. On the other hand, although LSDNet is a network with polarization characteristic data as input, it does not dig

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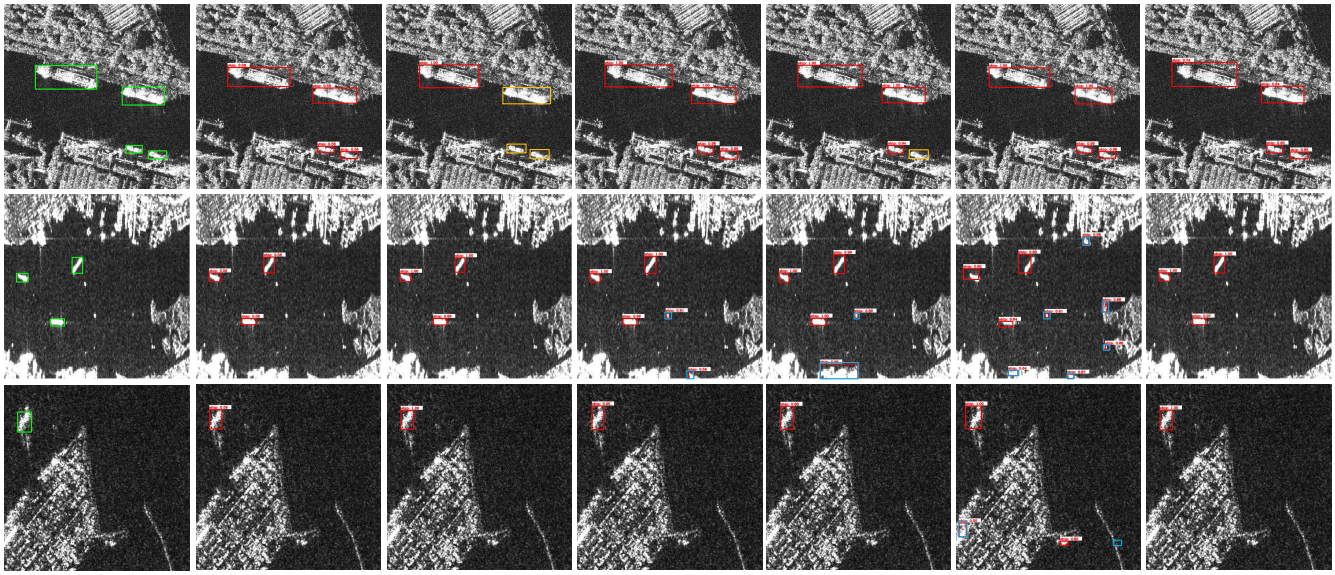


Fig. 27. Three representative inshore examples detection results in SSDD data. Green boxes indicate ground truth; red boxes show predicted results; yellow boxes denote missed detections; and blue boxes are false alarms. (a) Ground truth. (b) CenterNet. (c) SSD-512. (d) YOLOv4. (e) YOLOv4 + SE. (f) LSDNet. (g) SCANet.

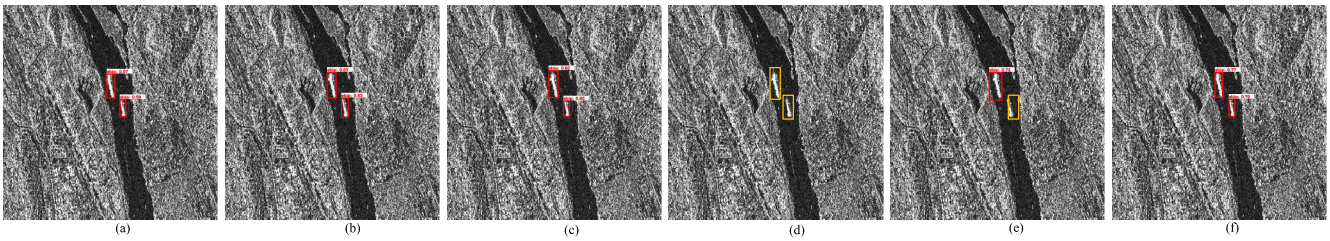


Fig. 28. Inland river ship detection results in the SSDD dataset. Green boxes indicate ground truth; red boxes denote predicted results; and yellow boxes mean missed detections. (a) CenterNet. (b) SSD-512. (c) YOLOv4. (d) YOLOv4 + SE. (e) LSDNet. (f) SCANet.

deeply into the polarization characteristics of the ship and thus cannot enrich the information of the ship. Although single-stage CenterNet, SSD-512, YOLOv4, and YOLOv4 + SE can effectively distinguish between ships and land targets, there were also many missed detections in the detection results. This is because they cannot fully explore the polarization features of the targets under different scattering mechanisms and cannot fuse the local and global features in the feature maps. In SCANet, SCAM is aware of the target polarization characteristics and fuses the local and global information of the feature map to mine the polarization characteristics of the target. As a result, SCANet can effectively improve the accuracy of target detection and has the best detection performance among the six networks.

5) Detection Results on the SSDD Dataset: To verify whether the proposed network has an equally good detection performance and generalization, six networks were tested on the publicly available SSDD dataset. The experimental results are listed in Table VIII. For the SSDD dataset, the highest Precision was achieved by CenterNet at 98.75%, followed by SCANet at 97.81%, whereas SSD-512, YOLOv4, YOLOv4 + SE, and LSDNet were all less precise than SCANet. SCANet had the highest Recall of 81.83%, which was 9.35%, 4.77%, 2.75%, 4.4%, and 6.05% higher than those of the other five networks, respectively. In addition, the *F1* score and AP values of SCANet are higher than those of the other five networks. Similarly, the detection results for these six networks were plotted as PR curves (Fig. 26). The curves corresponding to SACNet are located in the outermost part of Fig. 26, wrapping the other five networks. Therefore, the detection performance of SCANet on the SSDD dataset was better than that of the five networks.

In addition, we plotted the detection results, as shown in Fig. 27. The three inshore scenes are shown in Fig. 27. Scene 1 contained four inshore ships: CenterNet, YOLOv4, LSDNet, and SCANet successfully detected four ships; YOLOv4 + SE missed one ship; and SSD-512 missed three ships. In scenario 2, which contained three ships, CenterNet, SSD-512, and SCANet detected all ships, whereas YOLOv4, YOLOv4 + SE, and LSDNet generated 2, 2, and 6 false alarms, respectively, although they detected all ships. Scene 3 contains only one ship and a large land background, and only LSDNet generated two false alarms among the six networks. Fig. 28 shows the detection results of the six networks in the scenario with a large area covered by land, where CenterNet, SSD-512, YOLOv4, and SCANet have consistent detection results, detecting all ships; while YOLOv4 + SE and LSDNet miss two and one ships, respectively.

Fig. 29 shows the results of the six networks' detection in the offshore scene. As can be seen from Fig. 29, the detection performance of CenterNet drops sharply in the presence of very small-scale ships, as demonstrated in scenarios 2 and 3

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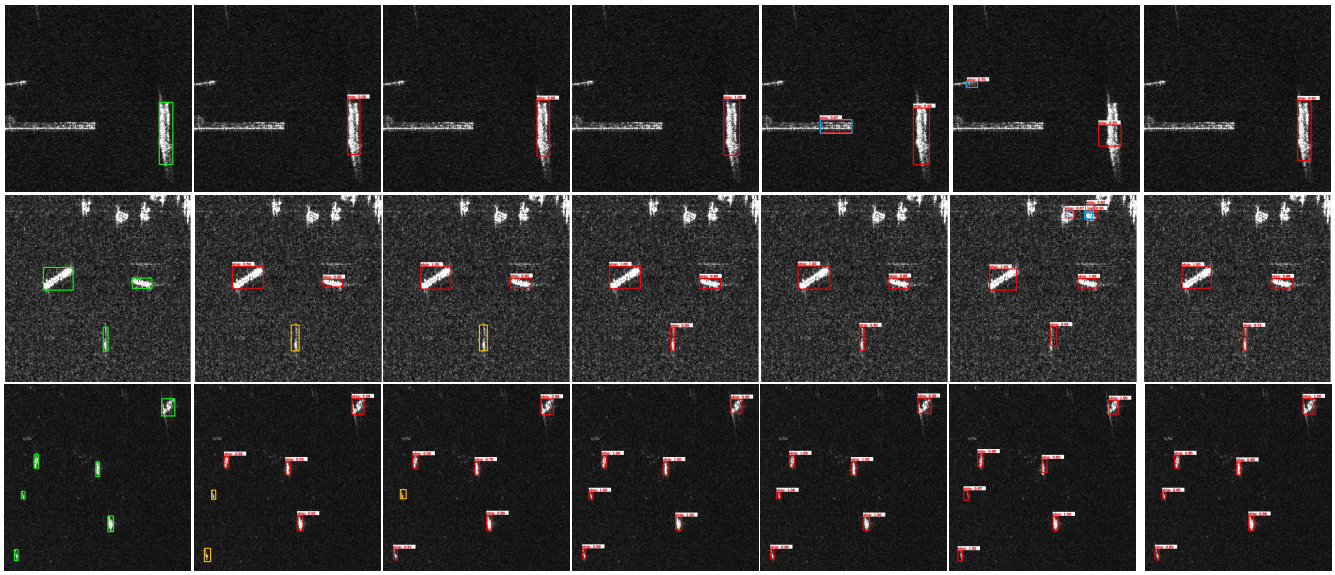


Fig. 29. Three representative offshore examples detection results in SSDD data. Green boxes indicate ground truth; red boxes show predicted results; yellow boxes denote missed detections; and blue boxes are false alarms. (a) Ground-truth. (b) CenterNet. (c) SSD-512. (d) YOLOv4. (e) YOLOv4 + SE. (f) LSDNet. (g) SCANet.

in Fig. 29. Similarly, the SSD-512 is equally insensitive to small-scale ships, resulting in missed detections. For YOV4 + SE and LSDNet, even if the scenes contained a small portion of land and the ships were far from the land, both still mistook the land as the target, resulting in false alarms. The SCANet proposed in this study detects all ships and has no false alarms.

Combining the four evaluation metrics and the detection results, we can conclude that SCANet has the best detection performance on the SSDD dataset and that all of its detection results are competitive. This is because the SCAM in SCANet can fully explore the information of ships between different channels and fuse them. In addition, it can effectively capture local and global features in the feature map, which significantly enriches ship information. Therefore, SCANet has outstanding detection results, regardless of whether it faces large-scale ships inshore or small-scale ships in the open sea. This also verifies that SCANet can maintain a good detection performance and generalization under different SAR datasets.

V. Conclusion

In this study, a four-component decomposition model named Ship-4SD is proposed to characterize ship scattering. The model is composed of Odd, Dbl, Od, and Asym components. In principle, for a ship, the Od component can describe the compounded scattering structure of odd- and even-bounced reflectors, while the asymmetric scattering structure can be characterized by the Asym component. To fully utilize the scattering components in the decomposition model, a SCANet, was designed by synthesizing the differences in scattering properties exhibited by different scattering components and the powerful feature extraction capability of CNNs.

According to both qualitative and quantitative evaluations, the rationality of the proposed Ship-4SD is confirmed; Ship-4SD is thus more applicable for characterizing the scattering of a ship than the existing four-component decomposition models, including BF4, G4U, Y4O, Y4R, and S4R. Furthermore, we compared Ship-4SD with the multicomponent models 6SD and 7SD and showed that Ship-4SD outperformed the multicomponent models in terms of both model decomposition efficiency and ship-sea differences. In summary, Ship-4SD is a suitable decomposition model for describing ship scattering structures. In addition, SCANet was verified to capture the contribution rate to ships under different scattering mechanisms to achieve the purpose of weakening the cluttered background and highlighting the target. Compared with other popular CNN-based ship detection networks, SCANet maintains the highest detection accuracy and the fewest false alarms. Moreover, SCANet provides good generalization for different target detection datasets.

In principle, there are similarities between ships and terrestrial buildings, and applying Ship-4SD to terrestrial urban areas may be potential. However, this requires validation using a large amount of measurement data. This investigation was limited to ship detection, and the analysis of terrestrial urban areas was beyond the scope of this study. In the future, we plan to explore this issue.

References

[1] R. Touzi, J. Hurley, and P. W. Vachon, "Optimization of the degree of polarization for enhanced ship detection using polarimetric RADARSAT-2," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5403–5424, Oct. 2015.

[2] X. Wang, G. Li, X.-P. Zhang, and Y. He, "A fast CFAR algorithm based on density-censoring operation for ship detection in SAR images," *IEEE Signal Process. Lett.*, vol. 28, pp. 1085–1089, 2021.

[3] H. Mahgoun, N. E. Chaffa, M. Ouarzeddine, and B. Souissi, "Application of polarimetric-SAR decompositions on RADARSAT-2 fine quad-Pol images to enhance the performances of ships detection algorithms," *Sens. Imag.*, vol. 21, no. 1, p. 56, Dec. 2020.

[4] M. Adil, F. Nunziata, A. Buono, D. Velotto, and M. Migliaccio, "Polarimetric scattering by a vessel at different incidence angles," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[5] L. Zhang, B. Zou, H. Cai, and Y. Zhang, "Multiple-component scattering model for polarimetric SAR image decomposition," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 603–607, Oct. 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:16:38 UTC from IEEE Xplore. Restrictions apply.

[6] R. Touzi and F. Charbonneau, "Characterization of target symmetric scattering using polarimetric SARs," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 11, pp. 2507–2516, Nov. 2002.

[7] J. R. Huynen, "Phenomenological theory of radar targets," Ph.D. dissertation, Tech. Univ. Delft, Delft, The Netherlands, 1970.

[8] W. An, Y. Cui, and J. Yang, "Three-component model-based decomposition for polarimetric SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 6, pp. 2732–2739, Jun. 2010.

[9] A. Freeman and S. L. Durden, "A three-component scattering model for polarimetric SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 3, pp. 963–973, May 1998.

[10] Y. Yamaguchi, T. Moriyama, M. Ishido, and H. Yamada, "Four-component scattering model for polarimetric SAR image decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 8, pp. 1699–1706, Aug. 2005.

[11] Y. Yamaguchi, A. Sato, W.-M. Boerner, R. Sato, and H. Yamada, "Four-component scattering power decomposition with rotation of coherency matrix," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2251–2258, Jun. 2011.

[12] A. Sato, Y. Yamaguchi, G. Singh, and S.-E. Park, "Four-component scattering power decomposition with extended volume scattering model," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 2, pp. 166–170, Mar. 2012.

[13] A. Bhattacharya, A. Muhuri, S. De, S. Manickam, and A. C. Frery, "Modifying the Yamaguchi four-component decomposition scattering powers using a stochastic distance," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3497–3506, Jul. 2015.

[14] G. Singh, Y. Yamaguchi, and S.-E. Park, "General four-component scattering power decomposition with unitary transformation of coherency matrix," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 3014–3022, May 2013.

[15] S.-W. Chen, X.-S. Wang, S.-P. Xiao, and M. Sato, "General polarimetric model-based decomposition for coherency matrix," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 3, pp. 1843–1855, Mar. 2014.

[16] G. Singh and Y. Yamaguchi, "Model-based six-component scattering matrix power decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5687–5704, Oct. 2018.

[17] G. Singh et al., "Seven-component scattering power decomposition of POLSAR coherency matrix," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8371–8382, Nov. 2019.

[18] B. Zou, Y. Zhang, N. Cao, and N. P. Minh, "A four-component decomposition model for PolSAR data using asymmetric scattering component," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 3, pp. 1051–1061, Mar. 2015.

[19] J. Li, C. Qu, and J. Shao, "Ship detection in SAR images based on an improved faster R-CNN," in *Proc. SAR Big Data Era, Models, Methods Appl. (BIGSARDATA)*, Nov. 2017, pp. 1–6.

[20] Z. Lin, K. Ji, X. Leng, and G. Kuang, "Squeeze and excitation rank faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 751–755, May 2019.

[21] Z. Cui, X. Wang, N. Liu, Z. Cao, and J. Yang, "Ship detection in large-scale SAR images via spatial shuffle-group enhance attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 379–391, Jan. 2021.

[22] C. Qin, X. Wang, G. Li, and Y. He, "A semi-soft label-guided network with self-distillation for SAR inshore ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5211814.

[23] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[24] T. Zhang et al., "LS-SSDD-v1.0: A deep learning dataset dedicated to small ship detection from large-scale Sentinel-1 SAR images," *Remote Sens.*, vol. 12, no. 18, p. 2997, Sep. 2020.

[25] L. Huang et al., "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 195–208, Jan. 2018.

[26] Q. Bai, G. Gao, X. Zhang, L. Yao, and C. Zhang, "LSDNet: Lightweight CNN model driven by PNF for PolSAR image ship detection," *IEEE J. Miniaturization Air Space Syst.*, vol. 3, no. 3, pp. 135–142, Sep. 2022.

[27] A. Guissard, "Mueller and Kennaugh matrices in radar polarimetry," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 3, pp. 590–597, May 1994.

[28] Y. Yamaguchi, Y. Yajima, and H. Yamada, "A four-component decomposition of POLSAR images based on the coherency matrix," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 3, pp. 292–296, Jul. 2006.

[29] T. Moriyama, "Polarimetric SAR image analysis using model fit for urban structures," *IEICE Trans. Commun.*, vol.88, no. 3, pp. 1234–1243, Mar. 2005.

[30] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[31] X. Ma, S. Hou, Y. Wang, J. Wang, and H. Wang, "Multiscale and dense ship detection in SAR images based on key-point estimation and attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5221111.

[32] Q. Hu, S. Hu, and S. Liu, "BANet: A balance attention network for anchor-free ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5222212.

[33] G. Margarit, J. J. Mallorqui, J. Fortuny-Guasch, and C. Lopez-Martinez, "Phenomenological vessel scattering study based on simulated inverse SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 4, pp. 1212–1223, Apr. 2009.

[34] V. Mnih, N. Heess, A. Graves, and K. Kavukcuoglu, "Recurrent models of visual attention," in *Proc. NIPS*. Cambridge, MA, USA: MIT Press, 2014, pp. 2204–2212.

[35] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[36] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[37] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 21–37.

[38] D. Li, Q. Liang, H. Liu, Q. Liu, H. Liu, and G. Liao, "A novel multidimensional domain deep learning network for SAR ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5203213.

[39] C. Zhang et al., "A novel full-polarization SAR image ship detector based on scattering mechanisms and wave polarization anisotropy," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 129–143, Aug. 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271622001630

[40] T. Zhang, Z. Yang, H. Gan, D. Xiang, S. Zhu, and J. Yang, "PolSAR ship detection using the joint polarimetric information," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8225–8241, Nov. 2020.

[41] X. Wang, D. Zhu, G. Li, X.-P. Zhang, and Y. He, "Proposal-copula-based fusion of spaceborne and airborne SAR images for ship target detection," *Inf. Fusion*, vol. 77, pp. 247–260, Jan. 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253521001585

[42] M. Adil, A. Buono, F. Nunziata, E. Ferrentino, D. Velotto, and M. Migliaccio, "On the effects of the incidence angle on the L-band multi-polarisation scattering of a small ship," *Remote Sens.*, vol. 14, no. 22, p. 5813, Nov. 2022.

[43] X. Yu et al., "The 1st tiny object detection challenge: Methods and results," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 315–323.

[44] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[45] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.



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